

AI-POWERED CUSTOMER SUPPORT & SERVICE AUTOMATION USING SALESFORCE SERVICE CLOUD & AGENTFORCE

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CHAPTER 1

INTRODUCTION

1.1 Project Overview

Customer support is one of the most critical functions in any electronics and gadgets company. Customers frequently contact support teams regarding product defects, warranty claims, order tracking, return requests, installation assistance, and general product inquiries. As the customer base grows, manually handling these requests becomes increasingly difficult, resulting in delayed responses, inconsistent customer experiences, and inefficient service management.

The **AI-Powered Customer Support & Service Automation** system is developed to overcome these challenges by automating customer service operations using **Salesforce Service Cloud**, **Salesforce Flows**, and **Agentforce (AI Service Agent)**. The system provides a centralized platform where customer information, product details, orders, cases, warranty claims, approvals, and support activities are managed efficiently.

The application automatically creates and routes customer support cases to appropriate support queues based on predefined business rules. Record-triggered Flows automate repetitive tasks such as email notifications, case assignment, status updates, and escalation processes, reducing manual intervention and improving operational efficiency.

Agentforce enhances the customer support experience by acting as an intelligent AI assistant capable of understanding customer queries and providing contextual responses. AI-generated recommendations help support agents resolve issues faster while maintaining consistency in service quality.

Managers can monitor customer service performance using interactive dashboards and reports that provide real-time insights into case volumes, resolution times, support workloads, SLA compliance, and customer satisfaction metrics.

Overall, the system transforms traditional customer support into a modern AI-driven service platform that improves customer satisfaction, increases agent productivity, and enables data-driven decision-making.

1.2 Problem Statement

An international Electronics & Gadgets company receives thousands of customer support requests daily through multiple communication channels. These requests commonly involve:

- Product malfunction and troubleshooting
- Warranty claim processing
- Order tracking requests
- Product replacement requests
- General customer inquiries
- Technical support

Currently, customer support operations rely heavily on manual processes. Support agents manually review incoming requests, assign cases, respond to customers, update records, and monitor service progress. This manual approach introduces several operational challenges.

Major issues include:

- Long customer waiting times
- Manual case assignment
- Repetitive support tasks
- Inconsistent customer responses
- Limited visibility into support performance
- Difficulty tracking Service Level Agreements (SLAs)
- Increased workload for support agents
- Delayed escalation of unresolved cases

As customer inquiries continue to increase, these problems directly affect customer satisfaction, operational efficiency, and business performance.

Therefore, an intelligent, automated, and scalable customer support solution is required to streamline service operations and improve the overall customer experience.

1.3 Proposed Solution

The proposed solution is an AI-powered customer support automation system developed using Salesforce Service Cloud and Agentforce.

The system integrates customer records, products, orders, warranty claims, support cases, automation flows, AI-powered assistance, reports, and dashboards into a single cloud-based platform.

The workflow begins when a customer submits a support request. Salesforce automatically creates a Case record and routes it to the appropriate support queue based on predefined assignment rules. Record-triggered Flows perform various automated actions such as:

- Sending acknowledgement emails
- Assigning support agents
- Updating case status
- Escalating overdue cases
- Sending warranty notifications
- Updating related records automatically

Agentforce assists both customers and support agents by generating intelligent responses, suggesting troubleshooting steps, answering frequently asked questions, and retrieving relevant knowledge articles.

Managers gain complete visibility into support operations through real-time reports and dashboards that display key performance indicators such as:

- Number of open cases
- Closed cases
- Average resolution time
- Pending warranty claims
- Queue workloads
- SLA compliance
- Customer satisfaction trends

The proposed system significantly reduces manual effort while improving response time, service quality, and operational efficiency.

Figure : Proposed Solution Workflow

Case Lifecycle Flow (Customer → Case → Queue → Agentforce → Resolution → Dashboard).

1.4 Objectives

The primary objective of this project is to design and implement an intelligent customer support management system that automates service operations using Salesforce technologies.

The specific objectives are:

- To automate customer case creation and management.
 - To maintain centralized records of customers, products, orders, and warranty claims.
 - To automate repetitive support activities using Salesforce Flows.
 - To implement intelligent case routing using support queues.
 - To integrate Agentforce AI for automated customer assistance.
 - To improve response time and case resolution efficiency.
 - To reduce manual workload for customer support agents.
 - To provide real-time dashboards for monitoring customer support performance.
 - To improve customer satisfaction through faster and more consistent service.
 - To build a scalable cloud-based customer service platform.
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1.5 Scope of the Project

The scope of the AI-Powered Customer Support & Service Automation system includes the complete lifecycle of customer service operations within an electronics and gadgets company.

The system provides functionality for:

- Customer Management
- Product Management
- Order Management
- Case Management
- Warranty Claim Management
- Queue-Based Case Assignment
- Approval Processes
- Email Notifications
- Salesforce Flow Automation
- Agentforce AI Integration
- Prompt Builder Integration
- Reports and Dashboards
- Lightning App Development
- Service Console Customization

The project focuses on automating customer support operations using Salesforce's declarative development tools without extensive custom coding. The system is designed to be scalable and can be extended in the future with additional communication channels such as live chat, WhatsApp integration, voice assistants, and predictive analytics.

1.6 Project Objectives at a Glance

Objective	Description
Improve Customer Experience	Reduce response time and improve service quality.
Automate Support Operations	Replace repetitive manual tasks with Salesforce Flows.
AI-Powered Assistance	Use Agentforce to provide intelligent responses and recommendations.
Centralized Data Management	Maintain customer, product, order, and warranty information in a single platform.
Efficient Case Management	Automatically create, assign, and resolve customer support cases.
Performance Monitoring	Generate reports and dashboards for service analysis.
Scalability	Support future business growth without major system changes.

1.7 Technologies Used

Layer	Technology
CRM Platform	Salesforce Service Cloud
User Interface	Salesforce Lightning Experience
Automation	Salesforce Flow
Artificial Intelligence	Agentforce (AI Service Agent)
Prompt Engineering	Prompt Builder
Database	Salesforce Cloud
Reporting	Salesforce Reports
Visualization	Salesforce Dashboards
Development Platform	Salesforce Developer Edition

2. IDEATION PHASE

2.1 Introduction

The ideation phase is one of the most important stages in the Software Development Life Cycle (SDLC). It focuses on understanding the business problem, identifying customer pain points, gathering stakeholder requirements, and proposing an effective solution before the actual development begins.

For an international Electronics & Gadgets company, customer support plays a significant role in maintaining customer satisfaction and brand reputation. As the company expands globally, the number of customer inquiries regarding product issues, warranty claims, order tracking, returns, and technical assistance continues to grow. Traditional manual customer support processes are no longer sufficient to efficiently handle these increasing requests.

The AI-Powered Customer Support & Service Automation system was conceptualized to eliminate repetitive manual tasks and improve service quality through automation and artificial intelligence. By leveraging Salesforce Service Cloud, Salesforce Flows, and Agentforce, the proposed solution provides intelligent case management, automated workflows, and AI-assisted customer interactions within a centralized platform.

This chapter presents the business problem, stakeholder analysis, customer journey, empathy map, brainstorming process, and proposed solution that guided the development of the system.

2.2 Business Problem

The company currently manages customer service requests manually. Customers contact the support team through email, phone calls, or web forms for various issues such as product defects, warranty registration, order tracking, and technical support.

Support agents manually review each request, create service records, assign cases, communicate with customers, and update the case status. As the number of customer requests increases, manual operations become inefficient and difficult to manage.

The major problems identified include:

- High volume of customer support requests.
- Delayed response and resolution times.
- Manual case assignment to support agents.
- Repetitive administrative tasks.
- Inconsistent responses from different agents.
- Lack of centralized customer information.
- Difficulty tracking warranty claims.
- Poor visibility into team performance.

- Delayed escalation of unresolved cases.
- Limited reporting and analytics.

These issues negatively affect customer satisfaction, increase operational costs, and reduce

2.3 Stakeholder Analysis

The success of the AI-Powered Customer Support & Service Automation system depends on several stakeholders who interact with the application throughout the customer service lifecycle.

Stakeholder	Role	Responsibilities
Customer	End User	Submit support requests, warranty claims, and order inquiries.
Support Agent	Service Representative	Handle customer cases, communicate with customers, and resolve issues.
Warranty Team	Specialized Support	Verify warranty eligibility and process warranty claims.
Service Manager	Supervisor	Monitor agent performance, assign workloads, and review reports.
Salesforce Administrator	System Administrator	Configure Salesforce objects, automation, users, permissions, and security.
AI Service Agent (Agentforce)	Virtual Assistant	Answer customer queries, recommend solutions, and assist support agents.

2.4 Customer Pain Points

The current customer support process creates several challenges for both customers and employees.

Customer Pain Points

- Long waiting time before receiving responses.
- Difficulty tracking order status.
- Delayed warranty approvals.
- Repeating the same issue multiple times.
- Inconsistent support experience.

- Lack of transparency in case progress.

Support Agent Pain Points

- Manual case creation.
- Large number of repetitive tasks.
- Difficulty prioritizing urgent cases.
- High workload.
- Searching multiple systems for customer information.
- Delayed communication between departments.

Manager Pain Points

- Limited visibility into case performance.
- Difficulty measuring agent productivity.
- Manual report preparation.
- No real-time dashboards.
- Poor SLA monitoring.

2.5 Empathy Map

Understanding user expectations helps in designing an efficient customer support system.

Customer Empathy Map

Thinks	Feels
"I need quick assistance."	Frustrated when responses are delayed.
"I hope my warranty is valid."	Worried about product issues.
"I want accurate updates."	Anxious while waiting for resolution.
Says	Does
"Where is my order?"	Contacts customer support.
"My product stopped working."	Creates support requests.
"Can I claim warranty?"	Uploads required documents.

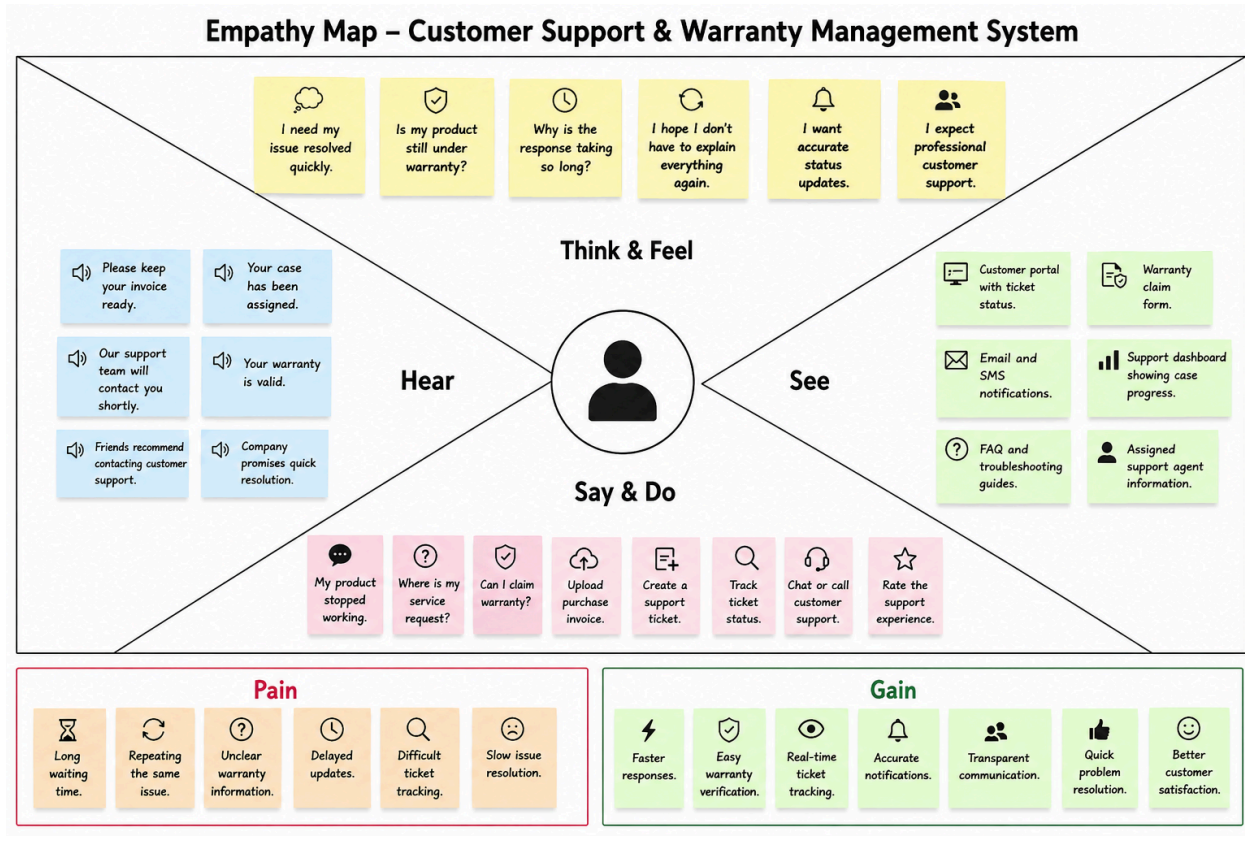
Support Agent Empathy Map

Thinks	Feels
"Too many repetitive tasks."	Overloaded with work.
"I need customer history quickly."	Pressured to meet SLAs.
Says	Does
"Let me check your case."	Updates case records.
"I'll escalate your issue."	Assigns cases and communicates with customers.

Service Manager Empathy Map

Thinks	Feels
"Are cases being resolved on time?"	Concerned about service quality.
"Which queue is overloaded?"	Responsible for team performance.
Says	Does
"Generate today's report."	Monitors dashboards and KPIs.
"Reassign high-priority cases."	Balances workloads.

Figure 2.2 Empathy Map



2.6 Brainstorming

Multiple approaches were considered before selecting Salesforce Service Cloud as the implementation platform.

Solution	Advantages	Limitations
Manual Email Support	Easy to start	Slow, no automation
Spreadsheet Tracking	Low cost	Difficult to maintain
Basic Ticketing Software	Centralized cases	Limited AI capabilities
Custom Web Application	Flexible	High development effort
Salesforce Service Cloud with Agentforce	Automation, AI, scalability, reporting	Learning curve

After evaluating these alternatives, Salesforce Service Cloud with Agentforce was selected because it provides enterprise-grade customer support capabilities with minimal custom coding.

2.7 Proposed Solution

The proposed solution integrates Salesforce Service Cloud, Agentforce, Salesforce Flow, and Prompt Builder into a unified customer support platform.

The solution performs the following operations:

- Stores customer information.
- Maintains product records.
- Tracks customer orders.
- Creates support cases.
- Processes warranty claims.
- Automatically assigns cases to support queues.
- Sends acknowledgement emails.
- Escalates overdue cases.
- Generates AI-powered responses.
- Provides dashboards for managers.
- Produces real-time reports.

The system minimizes manual effort while ensuring faster response times and improved customer satisfaction.

Proposed Solution Workflow

Customer → Case → Queue → Agentforce → Support Agent → Resolution → Dashboard.

2.8 Solution Benefits

The proposed solution provides significant improvements over the existing manual process.

Operational Benefits

- Automated case assignment.
- Faster issue resolution.

- Reduced manual workload.
- Consistent customer communication.
- AI-assisted responses.
- Centralized customer data.

Business Benefits

- Improved customer satisfaction.
- Increased agent productivity.
- Better SLA compliance.
- Enhanced reporting capabilities.
- Improved managerial decision-making.
- Reduced operational costs.

Technical Benefits

- Cloud-based architecture.
 - Scalable platform.
 - Secure data management.
 - Easy customization.
 - Low-code development.
 - Seamless Salesforce integration.
-

2.9 Feasibility Analysis

Technical Feasibility

Salesforce Service Cloud provides all the required features, including case management, workflow automation, reports, dashboards, AI integration, and security, making the project technically feasible.

Economic Feasibility

By reducing manual work, improving agent productivity, and decreasing customer resolution time, the system lowers operational costs and provides long-term business value.

Operational Feasibility

The Lightning interface is intuitive and user-friendly. Support agents, managers, and administrators can quickly adopt the system with minimal training.

Schedule Feasibility

Using Salesforce's low-code platform, the project can be completed within the planned academic timeline while meeting all functional requirements.

3.REQUIREMENT ANALYSIS PHASE

3.1 Introduction

Requirement Analysis is one of the most critical phases in the Software Development Life Cycle (SDLC). During this phase, the functional and non-functional requirements of the proposed system are identified, analyzed, and documented to ensure that the final solution satisfies business objectives and user expectations.

For the AI-Powered Customer Support & Service Automation system, requirement analysis focuses on understanding the operational needs of an international Electronics & Gadgets company that handles a large volume of customer support requests. The system must efficiently manage customer records, products, orders, support cases, warranty claims, approvals, queues, AI-assisted responses, and service analytics while reducing manual effort and improving customer satisfaction.

The requirements collected during this phase serve as the foundation for designing, implementing, testing, and maintaining the Salesforce Service Cloud application.

3.2 Functional Requirements

Functional requirements describe the features and operations that the system must perform.

Requirement ID	Functional Requirement	Description
FR-1	Customer Management	Create, update, search, and maintain customer information.
FR-2	Product Management	Maintain product catalog with product details and warranty information.
FR-3	Order Management	Store customer orders and provide order tracking information.

FR-4	Case Management	Create, assign, update, escalate, and close customer support cases.
FR-5	Warranty Claim Management	Register, verify, approve, and track warranty claims.
FR-6	Queue Management	Automatically assign cases to the appropriate support queues.
FR-7	Approval Process	Manage approvals for warranty claims and high-priority requests.
FR-8	Flow Automation	Automate notifications, assignments, escalations, and record updates using Salesforce Flows.
FR-9	AI Customer Support	Use Agentforce AI to answer customer queries and recommend solutions.
FR-10	Prompt Builder Integration	Generate intelligent responses using Prompt Builder templates.
FR-11	Email Notifications	Automatically notify customers regarding case updates and warranty status.
FR-12	Reports	Generate reports on support performance and case resolution.
FR-13	Dashboards	Display real-time service metrics using interactive dashboards.
FR-14	User Management	Manage users, roles, profiles, and permission sets.

3.3 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system.

NFR-1 Usability

The system should provide an intuitive and user-friendly interface so that support agents, managers, and administrators can perform their tasks efficiently without requiring extensive technical training.

Requirements

- Simple Lightning Interface
- Easy Navigation
- Responsive Forms
- Interactive Dashboards
- Quick Search Functionality

Example

A newly hired support agent should be able to log in, access customer cases, update case status, and resolve customer requests with minimal training.

NFR-2 Security

The application must ensure that customer information, warranty records, and support cases are protected from unauthorized access.

Requirements

- Role-Based Access Control
- Profiles and Permission Sets
- Object-Level Security
- Field-Level Security
- Secure Authentication
- Data Encryption

Example

Only Warranty Managers should have permission to approve warranty claims, while Support Agents can only create and update customer cases.

NFR-3 Reliability

The system should consistently perform customer service operations without data loss or unexpected failures.

Requirements

- Automatic Record Saving
- Reliable Cloud Storage
- Error Handling
- Data Recovery
- High Availability

Example

If a network interruption occurs while updating a customer case, the system should preserve previously saved information.

NFR-4 Performance

The application should respond quickly even when processing thousands of customer cases.

Requirements

- Fast Page Loading
- Efficient Search
- Quick Report Generation
- Minimal Processing Delay

Example

Managers should be able to generate support performance reports within a few seconds.

NFR-5 Availability

The customer support application should remain available whenever support teams need it.

Requirements

- 24×7 Accessibility
- Cloud-Based Infrastructure
- Disaster Recovery
- High System Uptime

Example

Support agents located in different countries should be able to access customer cases at any time.

NFR-6 Scalability

The application should accommodate increasing numbers of customers, products, and support requests without affecting system performance.

Requirements

- Support Additional Users
- Handle Large Volumes of Cases
- Expand Data Storage
- Maintain Performance

Example

As the electronics company expands globally, the system should efficiently support millions of customer records.

NFR-7 Maintainability

The system should be easy to update and maintain.

Requirements

- Low-Code Configuration
- Modular Components
- Reusable Flows
- Easy Customization

Example

Administrators should be able to modify business rules without changing program code.

3.4 User Requirements

The following table summarizes the requirements of different users interacting with the system.

User	Requirements
Customer	Submit cases, track orders, check warranty status, receive notifications.
Support Agent	View assigned cases, update case status, resolve issues, communicate with customers.
Warranty Team	Verify warranty claims and process approvals.
Service Manager	Monitor dashboards, analyze reports, manage queues, review SLA compliance.
Salesforce Administrator	Configure objects, automation, users, security, reports, and AI settings.
Agentforce AI	Provide automated responses, suggest solutions, and assist support agents.

3.5 System Workflow

The AI-Powered Customer Support & Service Automation system follows a structured workflow that automates the complete customer support lifecycle.

Workflow Steps

1. Customer submits a support request.
 2. Salesforce creates a Case record.
 3. Case Assignment Rules route the case to the appropriate Queue.
 4. Record-Triggered Flow sends an acknowledgement email.
 5. Agentforce analyzes the customer's issue.
 6. AI recommends possible solutions.
 7. Support Agent reviews and resolves the issue.
 8. Warranty Approval Process is initiated if required.
 9. Case status is updated automatically.
 10. Customer receives the final resolution.
 11. Reports and Dashboards update automatically.
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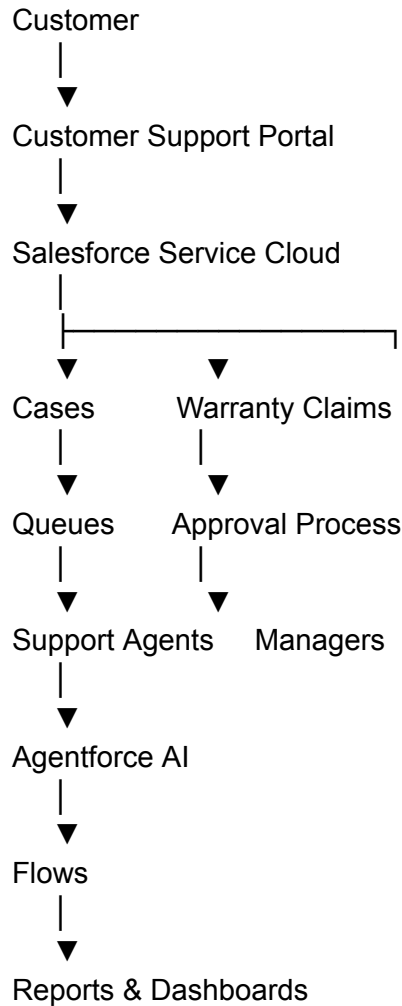
Figure 3.1 System Workflow

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Suggested Screenshot: Overall Customer Support Workflow or Flow Diagram.

3.6 Data Flow Diagram (DFD)

The Data Flow Diagram illustrates how information flows through the customer support system.



3.7 System Architecture

The application follows a layered architecture to ensure scalability, modularity, and maintainability.

Presentation Layer

Provides the Salesforce Lightning Experience interface where customers, support agents, managers, and administrators interact with the system.

Application Layer

Implements the business logic using:

- Salesforce Flows
 - Assignment Rules
 - Approval Processes
 - Validation Rules
 - Prompt Builder
 - Agentforce AI
-

Data Layer

Stores information using Salesforce Standard and Custom Objects.

Examples:

- Account
 - Contact
 - Product
 - Order
 - Case
 - Warranty Claim
-

AI Layer

Uses Agentforce AI to:

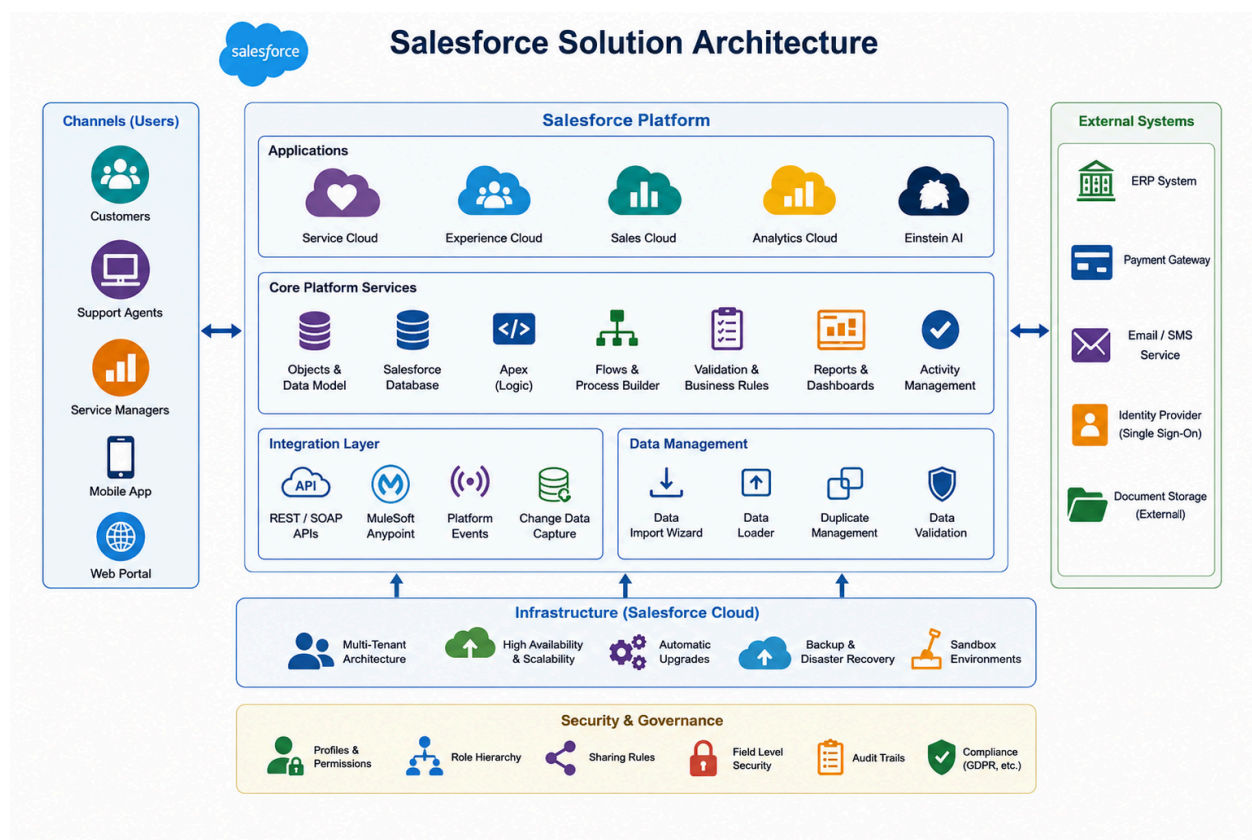
- Answer customer questions
 - Recommend troubleshooting steps
 - Retrieve knowledge articles
 - Generate intelligent responses
-

Analytics Layer

Provides business intelligence using:

- Reports
- Dashboards
- Charts
- Service Metrics
- KPIs

Figure Solution Architecture



3.8 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning Experience

CRM Platform	Salesforce Service Cloud
Database	Salesforce Cloud
Automation	Salesforce Flow
AI	Agentforce
Prompt Engineering	Prompt Builder
Security	Profiles, Roles, Permission Sets
Reporting	Salesforce Reports
Visualization	Salesforce Dashboards
Development Platform	Salesforce Developer Edition

3.9 Assumptions and Constraints

Assumptions

- Users have valid Salesforce credentials.
- Customer information is accurate.
- Internet connectivity is available.
- Service Cloud features are enabled.
- Agentforce is properly configured.

Constraints

- The project is developed using Salesforce's declarative tools.
- Internet access is required to use the application.
- AI responses depend on configured prompts and available knowledge.
- Some advanced features require Salesforce licensing.

4.PROJECT DESIGN PHASE

4.1 Introduction

The Project Design Phase transforms the analyzed business requirements into a structured solution architecture that serves as the blueprint for system implementation. During this phase, the overall system architecture, business processes, data model, automation strategy, user interactions, and AI integration are designed to ensure that the application meets both functional and non-functional requirements.

The **AI-Powered Customer Support & Service Automation** system is designed using **Salesforce Service Cloud, Salesforce Flows, Agentforce (AI Service Agent), and Prompt Builder**. The application follows Salesforce's low-code development approach, enabling rapid development while ensuring scalability, security, maintainability, and high performance.

The design focuses on automating customer support processes, reducing manual intervention, improving response times, and providing intelligent assistance through AI-powered recommendations.

4.2 Problem–Solution Fit

The existing customer support process relies heavily on manual operations, making it difficult to efficiently manage increasing customer requests. Manual case creation, assignment, follow-up, and reporting lead to delays, inconsistent service quality, and reduced operational visibility.

The proposed Salesforce-based solution addresses these challenges by automating repetitive tasks and integrating artificial intelligence into customer support operations.

Existing Problems

- Manual customer support operations
- Delayed response time
- Manual case assignment
- Repetitive administrative work
- Lack of centralized customer information
- Poor case tracking
- Slow warranty approval process

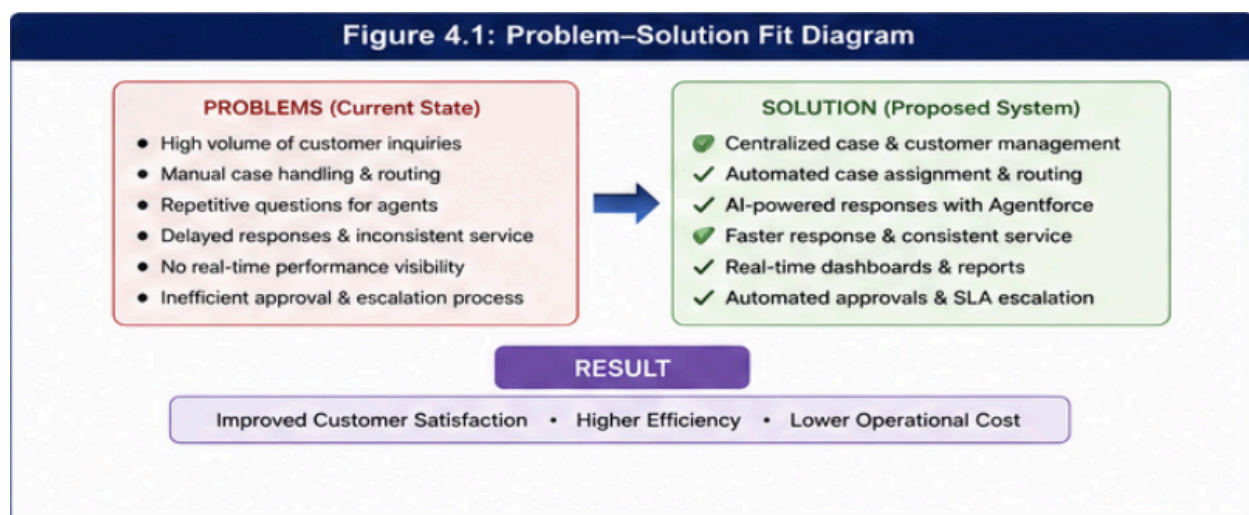
- Limited reporting capabilities
- Inconsistent customer responses

Proposed Solution

The proposed system introduces automation and AI to streamline customer support activities.

Existing Problem	Proposed Salesforce Solution
Manual case creation	Automatic Case Management
Manual assignment	Queue-Based Assignment Rules
Delayed responses	Automated Email Notifications
Repetitive work	Record-Triggered Flows
Customer inquiries	Agentforce AI Assistance
Warranty approvals	Automated Approval Processes
Manual reporting	Salesforce Reports & Dashboards
Poor visibility	Real-Time Analytics
Slow resolution	AI Recommendations & Automation

The proposed solution significantly improves operational efficiency while providing a faster and more consistent customer support experience.



4.3 Proposed Solution

The AI-Powered Customer Support & Service Automation system centralizes customer service operations within Salesforce Service Cloud. The system integrates customer information, product details, orders, warranty claims, support cases, automation flows, AI capabilities, and analytics into a single cloud platform.

Whenever a customer submits a support request, Salesforce automatically creates a Case record. Assignment Rules determine the appropriate support queue, while Record-Triggered Flows automate acknowledgements, notifications, and escalations. Agentforce analyzes customer queries and provides intelligent recommendations to support agents, helping them resolve issues more efficiently.

Managers can monitor service performance through real-time dashboards and reports, enabling better decision-making and resource allocation.

The proposed solution offers the following capabilities:

- Centralized customer data management
- Automated case lifecycle management
- Intelligent queue-based routing
- Warranty claim automation
- AI-assisted customer support
- Prompt-based response generation
- Automated email notifications
- SLA monitoring
- Reports and dashboards
- Secure access control



4.4 Solution Architecture

The system follows a layered architecture to ensure modularity, scalability, and maintainability.

Presentation Layer

Provides the user interface using Salesforce Lightning Experience.

Users include:

- Customers
- Support Agents
- Service Managers
- Salesforce Administrators

Functions:

- Case Management
 - Product Management
 - Warranty Claims
 - Reports
 - Dashboards
 - Service Console
-

Business Logic Layer

Implements business rules using Salesforce declarative tools.

Components include:

- Validation Rules
 - Assignment Rules
 - Approval Processes
 - Record Types
 - Page Layouts
 - Queues
-

Automation Layer

Handles process automation using Salesforce Flow.

Automated operations include:

- Case Assignment
 - Email Notifications
 - Case Escalation
 - Warranty Updates
 - Status Changes
 - Approval Notifications
-

AI Layer

Provides intelligent customer assistance using Agentforce.

Features:

- Customer Query Resolution
 - AI Recommendations
 - Prompt Builder Integration
 - Knowledge Retrieval
 - Suggested Responses
-

Data Layer

Stores application data using Salesforce Objects.

Standard Objects:

- Account
- Contact
- Product
- Order
- Case

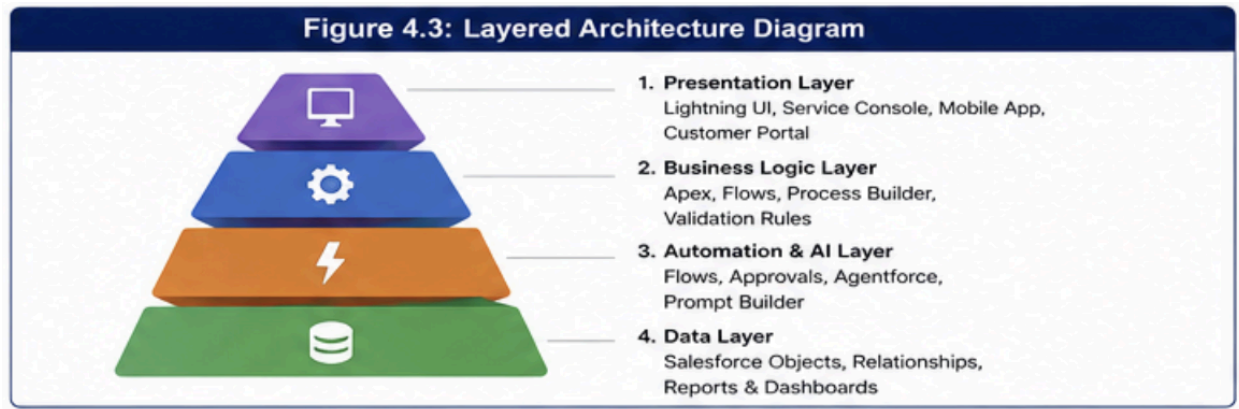
Custom Objects:

- Warranty Claim
-

Analytics Layer

Provides business intelligence using:

- Reports
- Dashboards
- Charts
- KPIs
- SLA Metrics

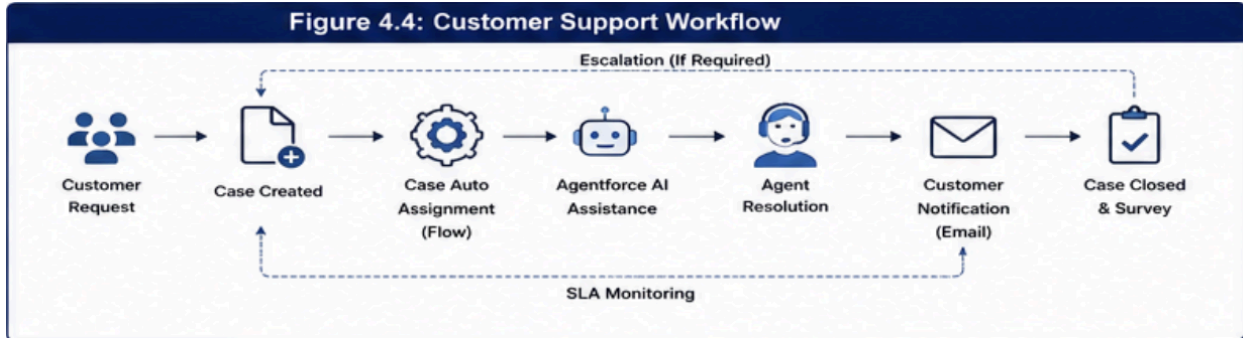


4.5 System Design Workflow

The customer support workflow is designed to automate the complete service lifecycle.

Workflow

1. Customer submits support request.
2. Case record is created.
3. Assignment Rules assign the case to the correct Queue.
4. Flow sends acknowledgement email.
5. Agentforce analyzes the issue.
6. AI suggests troubleshooting steps.
7. Support Agent reviews the case.
8. Warranty approval is initiated if necessary.
9. Case is resolved.
10. Customer receives resolution notification.
11. Reports and dashboards update automatically.



4.6 Object Design

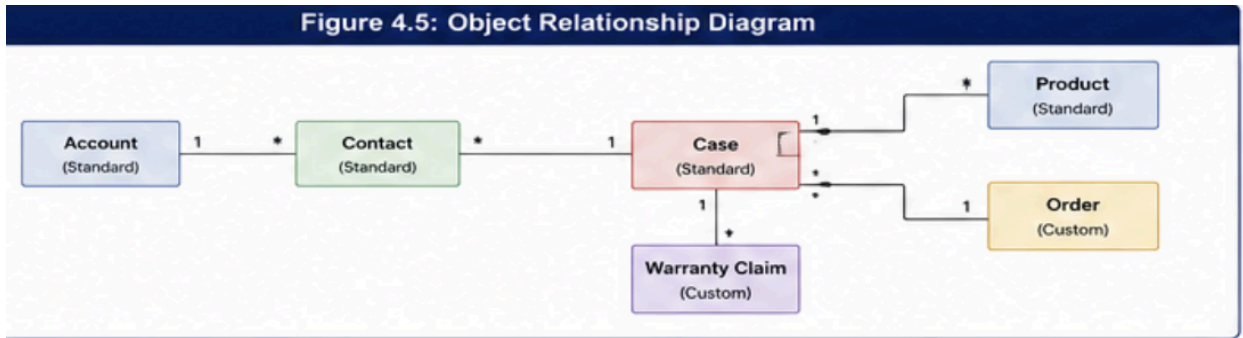
The application utilizes Salesforce Standard Objects along with one Custom Object.

Standard Objects

Object	Purpose
Account	Customer account information
Contact	Customer contact details
Product	Product catalog
Order	Customer order details
Case	Customer support requests

Custom Object

Object	Purpose
Warranty Claim	Stores warranty claim information and approval status



4.7 Automation Design

Automation is implemented using Salesforce Flow to eliminate repetitive manual tasks.

Implemented Flows

Flow	Purpose
Case Assignment Flow	Automatically assigns cases to support queues.
Acknowledgement Flow	Sends confirmation emails after case creation.
Warranty Notification Flow	Notifies customers about warranty status.
Escalation Flow	Escalates overdue cases to managers.
Case Closure Flow	Sends closure confirmation after resolution.

These automations improve consistency, reduce manual work, and ensure timely communication with customers.



4.8 AI Design

Artificial Intelligence is integrated using Agentforce.

AI Features

- Intelligent customer assistance
- Automated responses
- Troubleshooting recommendations
- Knowledge article suggestions
- Prompt Builder integration
- Agent productivity enhancement

The AI assistant reduces support workload while improving response accuracy and customer satisfaction.

4.9 Security Design

The application protects customer and business data using Salesforce security features.

Security Components

- Profiles
- Roles
- Permission Sets
- Field-Level Security
- Object-Level Security
- Login Authentication
- Sharing Rules

Only authorized users can access sensitive customer and warranty information.

4.10 Design Advantages

The proposed system offers several advantages:

- Modular architecture
 - Low-code development
 - AI-assisted customer support
 - Automated workflows
 - Centralized data management
 - Scalable cloud infrastructure
 - Improved customer satisfaction
 - Faster case resolution
 - Reduced manual effort
 - Real-time monitoring and analytics
-

5.PROJECT PLANNING & SCHEDULING

5.1 Introduction

Project Planning and Scheduling is an essential phase in the Software Development Life Cycle (SDLC) that ensures the successful execution of a software project within the defined scope, time, and available resources. It involves identifying project activities, allocating responsibilities, estimating timelines, managing potential risks, and monitoring project progress throughout the development lifecycle.

For the **AI-Powered Customer Support & Service Automation** project, Agile methodology was adopted because it provides flexibility, encourages continuous improvement, and supports incremental delivery of project features. The project was divided into multiple phases, allowing each module to be designed, developed, tested, and validated independently before integration.

Proper planning helped ensure that customer management, case management, warranty claims, Salesforce Flows, Agentforce integration, reports, dashboards, and Lightning application development were completed systematically within the project schedule.

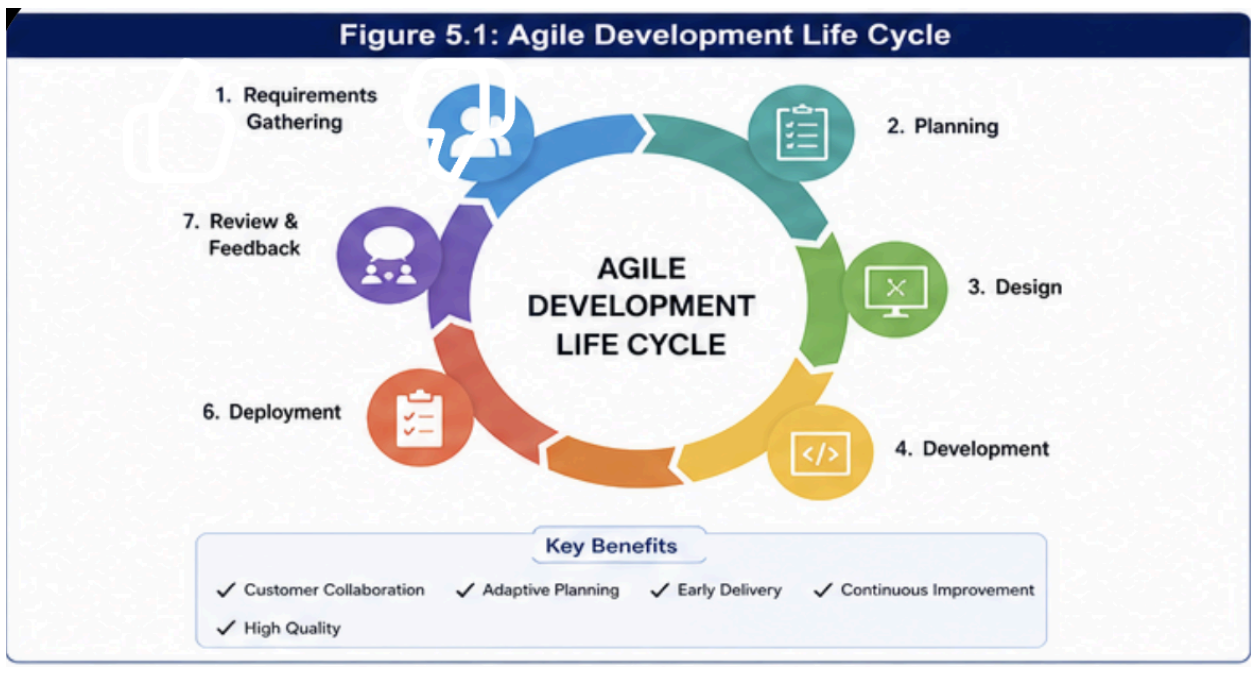
5.2 Project Development Methodology

The project follows the **Agile Software Development Methodology**, which emphasizes iterative development, collaboration, continuous feedback, and regular testing.

Unlike traditional Waterfall development, Agile enables developers to deliver working modules incrementally while accommodating changing business requirements.

Advantages of Agile Methodology

- Faster development cycles.
- Continuous testing and feedback.
- Easy adaptation to changing requirements.
- Better collaboration among stakeholders.
- Early identification of defects.
- Incremental feature delivery.
- Improved project quality.



5.3 Project Phases

The development of the AI-Powered Customer Support & Service Automation system was divided into the following phases.

Phase	Description

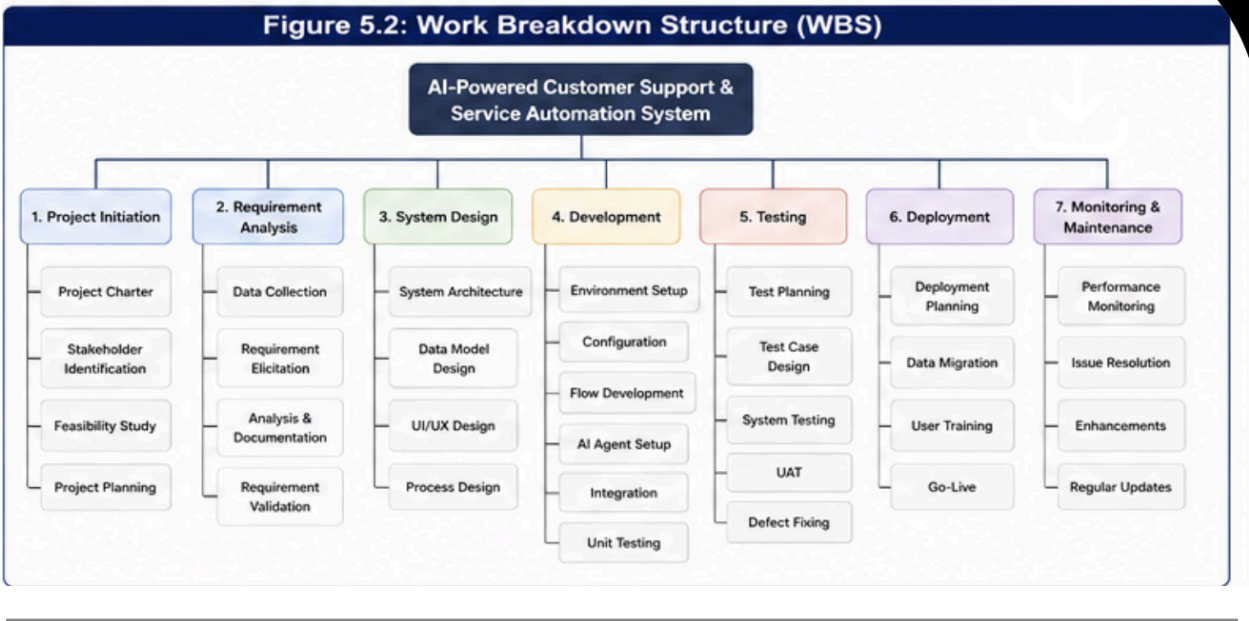
Requirement Gathering	Understanding customer support processes and business needs.
Requirement Analysis	Identifying functional and non-functional requirements.
Project Design	Designing system architecture, objects, workflows, and automation.
Salesforce Configuration	Creating objects, fields, page layouts, record types, and validation rules.
Flow Development	Implementing automation using Salesforce Flows.
Agentforce Integration	Configuring AI Service Agent and Prompt Builder.
Reports & Dashboards	Developing analytical reports and dashboards.
Testing	Performing functional and integration testing.
Deployment	Finalizing and presenting the application.

5.4 Work Breakdown Structure (WBS)

The project was divided into manageable tasks to ensure proper planning and execution.

Task ID	Task
T1	Requirement Gathering
T2	Requirement Analysis
T3	Salesforce Environment Setup
T4	Object Creation
T5	Field & Relationship Creation
T6	Validation Rules
T7	Queue Configuration
T8	Approval Process
T9	Salesforce Flow Development
T10	Lightning App Development

- T11 Agentforce Configuration
- T12 Prompt Builder Configuration
- T13 Reports Creation
- T14 Dashboard Development
- T15 Testing
- T16 Documentation
- T17 Final Deployment



5.5 Project Schedule

The project activities were scheduled to ensure timely completion while allowing adequate time for testing and documentation.

Week	Activities
Week 1	Requirement Gathering and Business Analysis
	Requirement Analysis and Project Design
	Salesforce Org Setup and Object Creation

Week 2	Fields, Relationships, Page Layouts, Record Types
	Validation Rules, Queues, Approval Processes
	Salesforce Flow Development
Week 3	Agentforce Configuration and Prompt Builder
	Reports and Dashboards
	Testing and Bug Fixing
	Documentation and Final Presentation

5.6 Sprint Planning

The project was executed using five Agile sprints.

Sprint	Duration	Objectives
Sprint 1	Week 1–2	Requirement Analysis and Project Design
Sprint 2		Salesforce Configuration (Objects, Fields, Layouts)
Sprint 3	Week 2–3	Automation using Flows and Approval Processes
Sprint 4		Agentforce, Prompt Builder, Reports & Dashboards
Sprint 5		Testing, Documentation, Deployment

Each sprint concluded with a review to verify completed work and plan the next iteration.

5.7 Resource Planning

The successful completion of the project required both software resources and human resources.

Software Resources

Resource	Purpose
Salesforce Developer Edition	Application Development
Salesforce Service Cloud	Customer Support Management
Agentforce	AI-Powered Customer Support
Prompt Builder	AI Prompt Templates
Visual Studio Code	Documentation and Development
Microsoft Word	Report Preparation
Draw.io	System Diagrams
Canva	Flowcharts and Architecture Diagrams

Human Resources

Role	Responsibility
Project Developer	Design and Implementation
Salesforce Administrator	Configuration and Security
Support Agent	User Testing
Service Manager	Requirement Validation
Project Guide	Review and Evaluation

5.8 Risk Management

Potential project risks were identified and mitigation strategies were prepared.

Risk	Impact	Mitigation Strategy
Requirement Changes	Medium	Frequent stakeholder reviews

Salesforce Configuration Errors	High	Regular testing after each configuration
Flow Automation Issues	Medium	Unit testing of each Flow
AI Response Inaccuracy	Medium	Optimize Prompt Builder templates
Internet Connectivity Issues	Medium	Cloud backup and reliable internet access
Project Delays	High	Weekly sprint reviews and milestone tracking

5.9 Milestone Planning

The project milestones were established to monitor development progress.

Milestone	Expected Outcome
M1	Requirement Analysis Completed
M2	Project Design Completed
M3	Salesforce Configuration Completed
M4	Automation Flows Completed
M5	Agentforce Integrated
M6	Reports and Dashboards Completed
M7	Testing Completed
M8	Documentation Completed
M9	Final Project Submission

5.10 Deliverables

At the end of the project, the following deliverables were produced:

- Salesforce Service Cloud Application
 - Customer Management Module
 - Product Management Module
 - Order Management Module
 - Warranty Claim Module
 - Case Management System
 - Queue Management
 - Approval Processes
 - Salesforce Flows
 - Agentforce AI Configuration
 - Prompt Builder Templates
 - Reports and Dashboards
 - Lightning App
 - Technical Documentation
 - Final Project Report
-

6.PROJECT DEVELOPMENT PHASE

6.1 Introduction

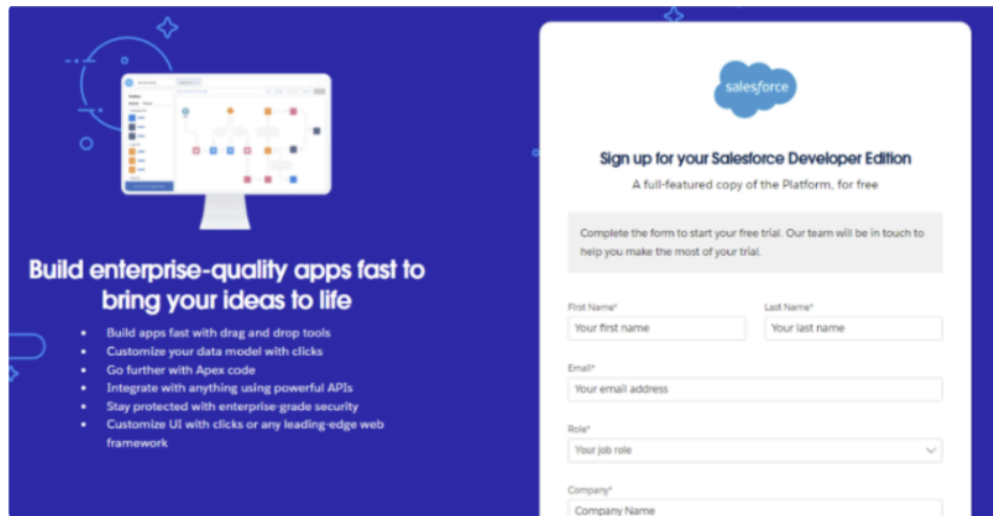
The Project Development Phase involves implementing the AI-Powered Customer Support & Service Automation system using Salesforce Service Cloud, Salesforce Flows, Agentforce, Prompt Builder, Reports, Dashboards, and Lightning Experience. Based on the requirements gathered during the analysis and design phases, the application was developed using Salesforce's declarative (low-code) tools, minimizing custom coding while ensuring scalability, maintainability, and security.

The development process includes configuring standard and custom objects, creating fields and relationships, designing page layouts and record types, implementing validation rules, configuring queues and approval processes, automating business processes using Salesforce Flows, integrating Agentforce for AI-assisted support, developing Prompt Builder templates, and creating reports and dashboards for monitoring customer service performance.

The following sections describe each development activity in detail.

Phase 1 Org Setup

Get your Trailhead Playground ready for Service Cloud



Step 1 — Log into your Trailhead Playground

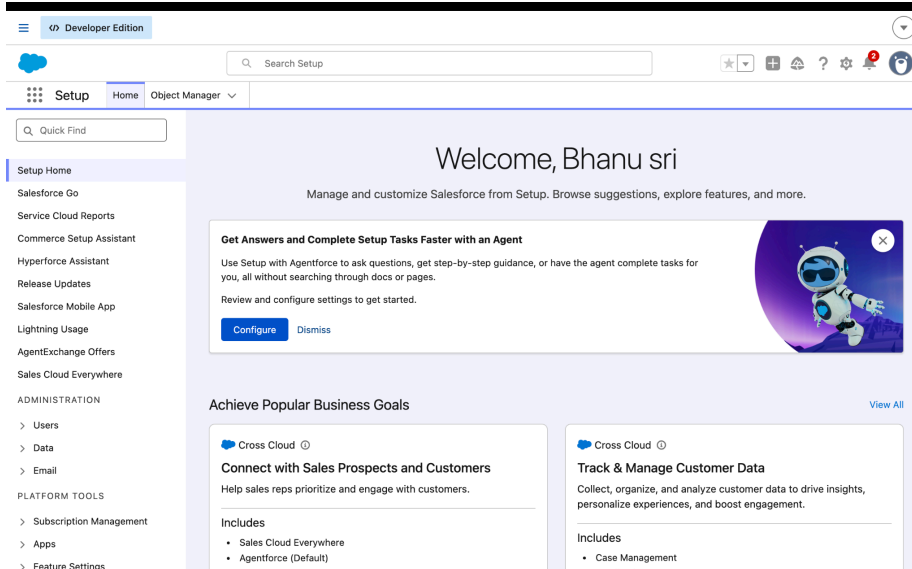
1. Go to **trailhead.salesforce.com**
 2. Click your **avatar / profile icon** (top right corner)
 3. Select **Hands-On Orgs** from the dropdown
 4. Click **Launch** next to your Trailhead Playground — Salesforce opens in a new tab
 5. Your org URL will look like: **https://xyz.lightning.force.com** — bookmark it
- **Tip:** If you don't have a Playground yet: go to *trailhead.salesforce.com* → *sign up free* → *Hands-On Orgs* → *Create Playground (takes 2–3 min)*

Step 2 — Open Setup

1. Inside Salesforce, click the **gear icon** ■ in the top-right corner
 2. Click **Setup** from the dropdown — Setup opens in a new tab
 3. You will use Setup constantly — keep this tab open throughout the project
 4. The **Quick Find box** (top-left in Setup) is your shortcut — type any feature name to jump to it directly
- **Tip:** Shortcut: in any Salesforce tab, type */setup* in the URL bar to open Setup instantly.

Step 3 — Enable Case Feed

1. In Setup Quick Find, type **Case Settings** → click it
2. Check the box for **Enable Case Feed** (if not already checked)
3. Click **Save**



Phase 2

Data Model

Add custom fields to Case + create the Warranty Claim object

■ ~30 min

Part A — Custom Fields on the Case Object

You will add 4 custom fields. All follow the same path: **Setup** → **Object Manager** → **Case** →

Fields &

Relationships → New

Step 1 — Add Case Type field (Picklist)

1. Setup → Object Manager → Case → Fields & Relationships → New

2. Field Type: select **Picklist** → click **Next**

3. Field Label: **Case Type** (API Name auto-fills as **Case_Type__c**)

4. Under Values, click **New** and add these one at a time: **Product Issue**, **Order Tracking**, **Warranty Claim**, **General Support**

5. Check **General Support** as the default value

6. Click **Next** → **Next** → **Save**

Step 2 — Add Resolution Notes field (Long Text Area)

1. Fields & Relationships → **New** → select **Long Text Area** → **Next**

2. Field Label: **Resolution Notes** (API Name: **Resolution_Notes__c**)

3. Length: **32768** | Visible Lines: **5**

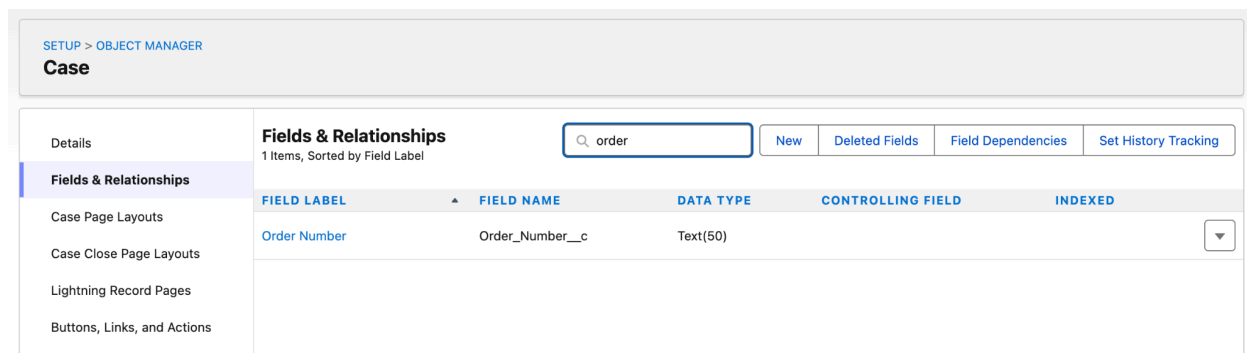
4. Click **Next** → **Next** → **Save**

Step 3 — Add Order Number field (Text)

1. Fields & Relationships → **New** → select **Text** → **Next**

2. Field Label: **Order Number** (API Name: **Order_Number__c**) | Length: **50**

3. Click **Next** → **Next** → **Save**



SETUP > OBJECT MANAGER

Case

Details

Fields & Relationships 1 Items, Sorted by Field Label

Search: order

New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Order Number	Order_Number__c	Text(50)		

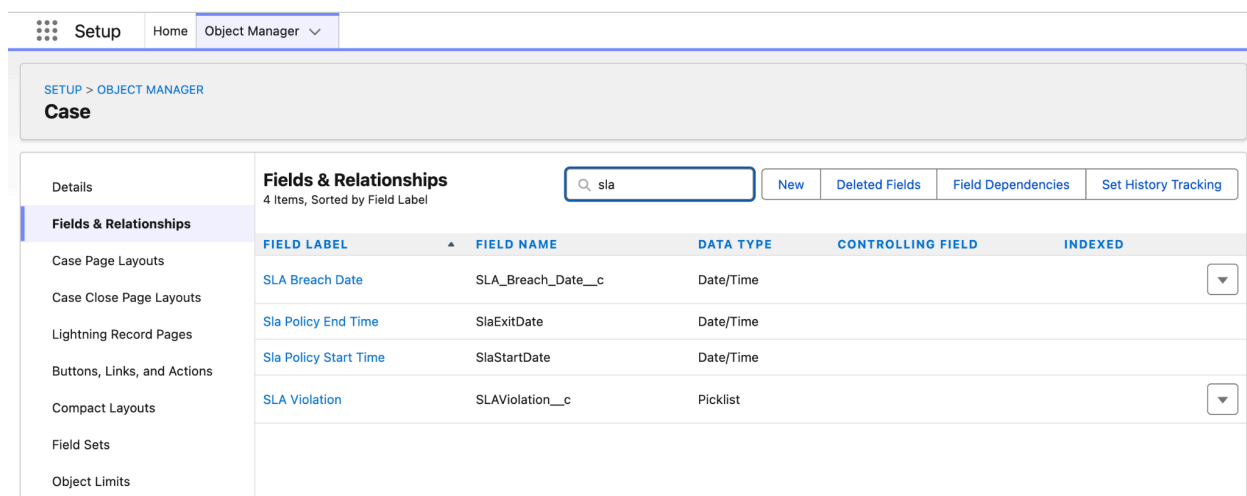
Step 4 — Add SLA Breach Date field (Date/Time)

1. Fields & Relationships → **New** → select **Date/Time** → **Next**

2. Field Label: **SLA Breach Date** (API Name: **SLA_Breach_Date__c**)

3. Click **Next** → **Next** → **Save**

■ **Tip:** You now have 4 custom fields on Case. Verify in Fields & Relationships — they show with API names ending in **__c**



Setup Home Object Manager

SETUP > OBJECT MANAGER

Case

Details

Fields & Relationships 4 Items, Sorted by Field Label

Search: sla

New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
SLA Breach Date	SLA_Breach_Date__c	Date/Time		
SLA Policy End Time	SLAExitDate	Date/Time		
SLA Policy Start Time	SLAStartDate	Date/Time		
SLA Violation	SLAViolation__c	Picklist		

Part B — Create the Warranty Claim Custom Object

Step 5 — Create the Warranty Claim object

1. Setup → Object Manager → top-right dropdown → **Create** → **Custom Object**

2. Label: **Warranty Claim** | Plural Label: **Warranty Claims** | Object Name: **Warranty_Claim**

3. Record Name type: **Auto Number** | Display Format: **WC-{0000}** | Starting Number: **1**

4. Check all three: *Allow Reports* ✓ | *Allow Activities* ✓ | *Track Field History* ✓

5. Click **Save**

Step 6 — Add 5 fields to Warranty Claim

1. Claim Date — Type: Date | Mark as Required → Save

2. Claim Status — Type: Picklist | Values: Submitted (default), Under Review, Approved, Rejected,

Closed $\rightarrow$ *Save*

3. Claim Amount — Type: Currency | Precision: 16, Scale: 2 → Save

4. Case — Type: Lookup Relationship | Related Object: **Case** → Save

5. Product — Type: *Lookup Relationship* | Related Object: **Product** → Save

■ **Tip:** For each field: Warranty Claim object → Fields & Relationships → New → choose type → fill details →

Next → *Next* → *Save*

Setup

Home

Object Manager ▾

SETUP > OBJECT MANAGER

Warranty Claim

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Scoping Rules

Object Access

Field Access

Fields & Relationships

9 Items, Sorted by Field Label

Q Quick Find

NewDeleted FieldsField DependenciesSet History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Case	Case__c	Lookup(Case)		✓
Claim Amount	Claim_Amount__c	Currency(16, 2)		
Claim Date	Claim_Date__c	Date		
Claim Status	Claim_Status__c	Picklist		
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Product	Product__c	Lookup(Product)		✓
Warranty Claim Name	Name	Auto Number		✓

Part C — Validation Rules

Step 7 — Validation Rule 1 — Require Resolution Notes on Close (on Case)

1. Object Manager → Case → Validation Rules → New

2. Rule Name: Require Resolution Notes On Close | Active: *checked*

3. Paste this formula in the Error Condition Formula box (see formula below):

4. Error Message: Resolution Notes are required before closing a case.

5. Error Location: *Field* → select *Resolution Notes*

6. Click Save

```
AND(
  ISPICKVAL(Status, "Closed"),
  ISBLANK(Resolution_Notes__c)
)
```

Step 8 — Validation Rule 2 — No Future Claim Date (on Warranty Claim)

1. Object Manager → **Warranty Claim** → **Validation Rules** → **New**

2. Rule Name: **No_Future_Claim_Date** | Active: **checked**

3. Error Condition Formula (paste exactly):

4. Error Message: **Claim Date cannot be in the future.**

5. Error Location: Field → **Claim Date** → Save

Claim_Date__c > TODAY()

Case Validation Rule

[Back to Case Validation Rules](#)

Validation Rule Detail

EditClone

Rule Name	Require_Resolution_Notes_On_Close	Active	✓
Error Condition Formula	AND(ISPICKVAL(Status, "Closed"), ISBLANK(Resolution_Notes__c))		
Error Message	Resolution Notes are required before closing a case.	Error Location	Resolution Notes
Description			
Created By	Bhanu sri Inturi , 6/26/2026, 11:18 PM	Modified By	Bhanu sri Inturi , 6/26/2026, 11:18 PM

EditClone

Phase 3

Flows

Build 3 record-triggered automation flows in Flow Builder

■ ~40 min

All 3 flows are Record-Triggered Flows. Path: **Setup** → **Quick Find: Flows** → **New Flow** →

Record-Triggered

Flow

Flow 1 — Case Auto Assignment

Step 1 — Create the Case Auto Assignment Flow

1. Setup → Quick Find → **Flows** → click **New Flow**

2. Select **Record-Triggered Flow** → click **Create**

3. Object: **Case** | Trigger: **A record is created** | Optimize for: **Actions and Related Records** → Done

Flow

Case Auto Assignment Flow

Open Flow

Type

Record—Run After Save

Associated Record

Element Error Rate

Details

Versions

Usage

Information

Flow Label

Case Auto Assignment Flow

API Name

Case_Auto_Assignment_Flow

Description

Flow Type

Record-Triggered After Save Flow

Associated Record

Segment

Created By

Bhanu.sri.Inturi, 6/26/2026, 11:56 PM

Created Date

6/26/2026, 11:56 PM

Last Modified

Bhanu.sri.Inturi, 6/26/2026, 11:56 PM

Last Modified Date

6/26/2026, 11:56 PM

4. Click the **+** button below Start → choose **Decision**
 5. Label the Decision: **Check Case Type**
 6. Add Outcome 1 — Label: **Product Issue** | Condition: **{!\$Record.Case_Type__c} Equals Product Issue**
 7. Add Outcome 2 — Label: **Order Tracking** | Condition: **{!\$Record.Case_Type__c} Equals Order Tracking**
 8. Add Outcome 3 — Label: **Warranty Claim** | Condition: **{!\$Record.Case_Type__c} Equals Warranty Claim**
 9. Default Outcome label: **General Support**
 - Step 2 — Add Update Records after each outcome branch
 1. Click **+** after the Product Issue outcome → Add **Update Records**
 2. How to find records: **Use the Case record that triggered the flow**
 3. Field to set: **Owner ID** | Value: paste the **Product Issues Queue ID** (get from Setup → Queues after Phase 4)
 4. Repeat for Order Tracking → **Order Support Queue ID**
 5. Repeat for Warranty Claim → **Warranty Queue ID**
 6. Repeat for Default → **General Support Queue ID**
 7. Click **Save** → Name: **Case Auto Assignment Flow** → click **Activate**
- **Note:** Come back after Phase 4 (Queues) to fill in the actual Queue IDs. The flow can be saved without IDs and edited later.

Developer Edition

Welcome

Flows

Recently Viewed

New

3 items • Updated a few seconds ago

Search this list...

⚙

↺

↻

✎

🗑

⌵

	Flow Label	Flow Type	E...	Status	A...	D...	Last Modified Date	Last ...	
1	Case Escalation Flow	Record-Triggered After Save Flow		Activated			6/27/2026, 12:11 AM	bha	⌵
2	Case Acknowledgment Email ...	Record-Triggered After Save Flow		Activated			6/27/2026, 12:03 AM	bha	⌵
3	Case Auto Assignment Flow	Record-Triggered After Save Flow		Activated			6/26/2026, 11:56 PM	bha	⌵

Flow 2 — Customer Acknowledgment Email

Step 3 — Create the Acknowledgment Email Flow

1. Flows → New Flow → Record-Triggered Flow
2. Object: Case | Trigger: A record is created
3. Add Entry Condition: ContactId Is Not Null (only send if a customer contact exists) → Done
4. Click + → Add element → Action → search Send Email → select it
5. To Email: {{\$Record.Contact.Email}}
6. Subject: We received your request — Case {{\$Record.CaseNumber}}
7. Body: type a friendly message using merge fields: {{\$Record.Contact.FirstName}}, {{\$Record.CaseNumber}}, {{\$Record.Case_Type__c}}
8. Click Done → Save → Name: Case Acknowledgment Email Flow → Activate
- INSERT SCREENSHOT HERE

Screenshot 3.2 — Flow Canvas: Acknowledgment Email

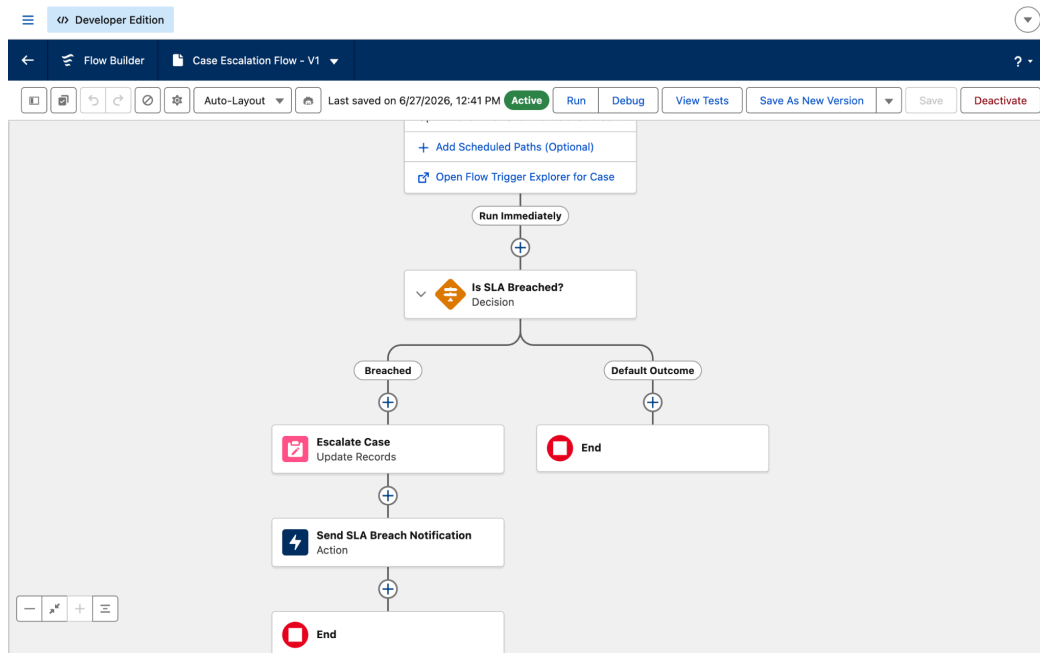
Show: Start (entry condition: ContactId not null) → Send Email action

Screenshot 3.2 — Flow Canvas: Acknowledgment Email

Flow 3 — Case Escalation on SLA Breach

Step 4 — Create the Escalation Flow

1. Flows → New Flow → Record-Triggered Flow
2. Object: Case | Trigger: A record is created or updated
3. Entry Conditions: Status Does Not Equal Closed AND SLA_Breach_Date__c Is Not Null → Done
4. Add Decision element → Label: Is SLA Breached?
5. Outcome condition: SLA_Breach_Date__c Less Than Or Equal To {{\$Flow.CurrentDateTime}} AND Priority Does Not Equal High
6. If breached → Update Records: set Priority = High, set IsEscalated = True
7. Then add Send Email action → To: {{\$Record.Owner:User.Email}} → Subject: SLA Breach Alert: Case {{\$Record.CaseNumber}}
8. Click Save → Name: Case Escalation Flow → Activate



Phase 4

Queues

Create 4 case routing queues and update the Assignment Flow

■ ~15 min

Step 1 — Create all 4 queues

1. Setup → Quick Find → **Queues** → click **New**
2. Label: **Product Issues Queue** | Queue Email: your email address
3. Under **Supported Objects**: click Add → select **Case** → Add
4. Under **Queue Members**: click Add → select your user → Add
5. Click **Save**
6. Repeat and create: **Order Support Queue, Warranty Queue, General Support Queue**

Step 2 — Copy the Queue IDs

1. After saving each queue, click on it to open the queue detail page
2. Look at the **browser URL bar** — copy the 18-character ID (starts with 00G...)
3. Save all 4 IDs somewhere (Notepad is fine) — you need them in the next step
4. Example URL: `.../lightning/setup/Queues/page?address=%2F00G...`

■ **Tip:** The Queue ID is the long alphanumeric code in the URL after the last forward slash before any ? symbol.

Queue

General Support Queue

[Help for this Page](#) 

[Edit](#) [Delete](#)

Label	General Support Queue	Queue Name	General_Support_Queue
Queue Email	bhanu.inturi7@gmail.com	Send Email to Members	☐
Grant Access Using Hierarchies	☐	Routing Configuration	
Supported Objects	Case		
Queue Description			
Created By	Bhanu sri Inturi, 6/26/2026, 11:37 PM	Modified By	Bhanu sri Inturi, 6/26/2026, 11:37 PM

[View All Users](#)

Name	Type
Bhanu sri Inturi	User

Step 3 — Update the Case Auto Assignment Flow with Queue IDs

1. Setup → Flows → open **Case Auto Assignment Flow**
2. Click **Edit** (you may need to deactivate it first → Deactivate button)
3. Click each **Update Records** element → change the OwnerId value from the placeholder to the real Queue ID
4. Update all 4 branches with their correct Queue IDs
5. Click **Save** → click **Activate**

Step 4 — Test the auto assignment

1. Go to App Launcher (9 dots) → **Cases** → click **New**
 2. Fill in Subject, set **Case Type = Product Issue** → Save
 3. Open the case → check the **Case Owner** field — it should show **Product Issues Queue**
 4. Create another case with Type = Order Tracking → verify it goes to Order Support Queue
- **Tip:** If the owner doesn't change, check that the flow is Active and the Queue IDs are correctly entered.

Queues

Queues allow groups of users to manage a shared workload more effectively. A queue is a location where records can be routed to await processing by a group member. The records remain in the queue until a user accepts them for processing or they are transferred to another queue. You can specify the set of objects that are supported by each queue, as well as the set of users that are allowed to retrieve records from the queue.

View: [All](#) [Edit](#) [Create New View](#)

Action	Label	Queue Name	Queue Email	Supported Objects	Modified By	Last Modified Date
Edit Del	General Support Queue	General_Support_Queue	bhanu.inturi7@gmail.com	Case	Inturi, Bhanu.sri	6/26/2026, 11:37 PM
Edit Del	Order Support Queue	Order_Support_Queue	bhanu.inturi7@gmail.com	Case	Inturi, Bhanu.sri	6/26/2026, 11:35 PM
Edit Del	Product Issues Queue	Product_Issues_Queue	bhanu.inturi7@gmail.com	Case	Inturi, Bhanu.sri	6/26/2026, 11:32 PM
Edit Del	Warranty Queue	Warranty_Queue	bhanu.inturi7@gmail.com	Case	Inturi, Bhanu.sri	6/26/2026, 11:36 PM

Phase 5

UI & Service Console

Build the Lightning app and customise the Case record page

■ ~25 min

Step 1 — Create the Service Console Lightning App

1. Setup → Quick Find → **App Manager** → click **New Lightning App**
 2. App Name: **Customer Support Console** | Developer Name: *auto-fills* → Next
 3. App Options: Navigation Style = **Console Navigation** → Next
 4. Utility Bar: click **Add Utility Item** → add **History** → Next
 5. Navigation Items: Add in this order — **Cases, Contacts, Accounts, Products, Warranty Claims, Reports, Dashboards** → Next
 6. User Profiles: Add **System Administrator** → Save & Finish
- **Tip:** To open the app: App Launcher (9 dots) → search Customer Support Console → click it. You'll see all navigation tabs at the top.

Desktop

Shrink To View

Refresh

Size

Activat

Close

Settings

Performance inadequate for second consecutive week

Follow Edit Delete Change Owner

Case Owner

Case Number

Contact Name

Account Name

Type

Case Reason

Case Type

Claim Amount

Case

Order Number

SLA Breach Date

Web Email

Web Name

Date/Time Opened

Product

Potential Liability

Created By

Subject

Description

Internal Comments

Resolution Notes

Status

Priority

Contact Phone

Contact Email

Case Origin

Web Company

Web Phone

Date/Time Closed

Engineering Req Number

SLA Violation

Last Modified By

Closed

High

(713) 842-3433

mycompany.com

Phone

Web Company

Web Phone

4/12/2016, 11:59 AM

76786

No

mycompany.com 4/12/2016, 9:16 AM

Performance inadequate for second consecutive week

Warranty Claims (0)

Filter: All Items - All Activities - All Types

Refresh Expand All View All

Showing < Overview

No activities to show

Get started by creating an event, scheduling a task, and more.

To change what's shown, try changing your filters.

View All Activities

Page > Activities

The Activities Timeline

The Activities component shows u
Activities View is set to Activity Tim
or user. [Tell Me More](#)

The Activities component expands
a scrollable list. Place it at the bott
it doesn't hide other components.

☐ Use tabbed activity view

Set Component Visibi

Filters

+ Add Filter

Step 2 — Create the Warranty Claims Tab (required for navigation)

1. Setup → Quick Find → **Tabs**
2. Under **Custom Object Tabs** → click **New**
3. Object: **Warranty Claim** | Tab Style: choose any icon (e.g. Shield)
4. Click **Next** → **Next** → **Save**
5. The Warranty Claims tab is now available to add to apps

Step 3 — Customise the Case Page Layout

1. Setup → Object Manager → **Case** → **Page Layouts** → click **Case Layout**
2. In the fields palette at the top, find **Case Type** → drag it to the Information section
3. Find **Order Number** → drag it next to Case Type
4. Find **SLA Breach Date** → drag it to the Information section
5. Find **Resolution Notes** → drag it to the bottom section (full width)
6. Click **Save**

Step 4 — Build a Custom Lightning Record Page for Case

1. Setup → Quick Find → **Lightning App Builder** → click **New**
2. Type: **Record Page** | Object: **Case** | Name: **Case Service Page** → **Next**
3. Template: select **Header and Two Columns** → **Done**
4. Drag **Highlights Panel** component to the top strip

5. Drag **Record Detail** to the left/main column
6. Drag **Related List** — **Single** to the right sidebar → **Properties: Related List = Warranty Claims**
7. Drag **Activity** below the related list
8. Click **Save** → **Activate** → Assign as **Org Default** → **Save**

Developer Edition

Search Setup

Setup Home Object Manager

app

Salesforce Mobile App

Data

Mass Transfer Approval Requests

Apps

App Manager

AppExchange Marketplace

Connected Apps

Connected Apps OAuth Usage

Manage Connected Apps

External Client Apps

External Client App Manager

OAuth Usage

Settings

Lightning Experience App Manager

New Lightning App New External Client App

32 items • Sorted by App Name • Filtered by All app menu items - TabSet Type, App Type

	App Name ↑	Developer Name	Description	Last Modified ...	App Type	Vi...
1	Agentforce Grid	AIWorkbench	Access Agentforce Grid	6/17/2026, 9:23 AM	Lightning	✓
2	Agentforce Studio	AgentforceStudio	Agentforce Studio	6/17/2026, 9:23 AM	Lightning	✓
3	All Tabs	AllTabSet		6/17/2026, 9:18 AM	Classic	
4	Analytics Studio	Insights	Build CRM Analytics dash...	6/17/2026, 9:18 AM	Classic	✓
5	App Launcher	AppLauncher	App Launcher tabs	6/17/2026, 9:18 AM	Classic	✓
6	Approvals	Approvals	Manage approvals and ap...	6/17/2026, 9:18 AM	Lightning	✓
7	Automation	FlowsApp	Automate business proce...	6/17/2026, 9:22 AM	Lightning	✓
8	Bolt Solutions	LightningBolt	Discover and manage bus...	6/17/2026, 9:18 AM	Lightning	✓
9	Community	Community	Salesforce CRM Communi...	6/17/2026, 9:18 AM	Classic	✓
10	Content	Content	Salesforce CRM Content	6/17/2026, 9:18 AM	Classic	✓
11	Customer Support Conso...	Customer_Support_Cons...		6/27/2026, 12:47 A...	Lightning	✓

Phase 6 Agentforce

■ ~20 min

Configure the AI Service Agent with 4 topics + Prompt Builder

Step 1 — Enable and create the AI Agent

1. Setup → Quick Find → **Agentforce**

2. If Agentforce is not visible: Setup → Quick Find → **Einstein Setup** → enable Einstein features first

3. Click **New Agent**

4. Agent Name: **Customer Support Agent** | Developer Name: **Customer_Support_Agent**

5. For Agent's User Record: click **Select User** → search for **bhanu** (your admin user) → select it

6. Click **Let's Go**

Developer Edition

Star

+

🏠

?

⚙️

🔔

👤

Search Setup

Setup

Home

Object Manager

Agentforce

Feature Settings

Commerce

Commerce Agentforce Settings

Einstein

Agentforce Data Library

Agentforce Gateway

Policies

Agentforce Registry

APIs

MCP Servers

Einstein Generative AI

Agentforce Studio

Agentforce Agents

Agentforce Agents

SETUP > AGENTFORCE STUDIO

Agentforce ☒ On

+ New Agent

Try the new Agentforce Builder!

The new Agentforce Builder is our AI-first experience for building, testing, and deploying enterprise-grade agents in Agent Script for more consistent behavior.

Let's Go

←

Agent Creator

New Agent

?

Help

BackNext

Steps

✓ Select an agent

✓ Select your subagents

3 Customize your agent

4 Select data sources

Name

Customer Support Agent

API Name

Customer_Support_Agent

* Description ⓘ

AI agent for customer support inquiries

961 characters remaining

* Role ⓘ

An AI customer service agent whose job is to help customers with support questions or other issues.

156 characters remaining

* Company ⓘ

Example: Your company specializes in providing customer relationship management (CRM) software.

Complete this field.

* Agent User ⓘ

New Agent User

×

☐ Keep a record of conversations with enhanced event logs to review agent behavior.

Learn More

Best Practices for Agent Settings

Agentforce Service Agent settings determine how an AI agent behaves and presents itself in conversations. Use these best practices to help you write concise and conversational descriptions.

Description

Describe the primary goals of the agent, including how it should achieve those goals. Include details that help the AI agent understand its end users, such as customers' context, preferences, and common behaviors.

Role

The role setting is the job description for the agent – it tells the agent what role it's playing in your company. Include key responsibilities, functions, and the target audience. Describe key tasks that it performs on a day-to-day basis as well as specialized or one-off tasks. Start the role description with "You are..".

Company

Tell the agent about the company it represents. Describe what your company does, who its target customers are, and the value proposition of important products or services you sell. Add details about what

■ **Note:** If you see a 'user limit' error: this is a known Trailhead Playground limitation. See the note box below for the fix.

AGENTFORCE LICENSE LIMITATION FIX:

Option A (Recommended): In Select User search, type your own admin username and select it.

Option B: Setup → Users → Edit 'Bryant, Victoria' → uncheck Active → Save.
Then try creating the agent again with your admin user.

Option C: If neither works, take a screenshot of the setup screen, document the 4 topic

instructions below, and note the limitation in your submission. This is accepted.

Step 2 — Add Topic 1 — Order Tracking

1. Inside your agent → click **Add Topic**

2. Topic Label: **Order Tracking**

3. Paste the instructions below into the Instructions field

4. Click **Save**

When a customer asks about their order:

1. Ask for their order number or registered email address
2. Look up the related Case or Order record in Salesforce
3. Return the current order status, expected delivery date, and any shipping notes
4. If the order has a problem (delayed, missing, damaged), offer to create a support case
5. Always confirm the customer's identity before sharing any order information

Create a Subagent

* Name: Order Tracking and Resolution

* API Name: Order_Tracking_and_Resolution

* Classification Description: Assist customers with tracking their orders, checking delivery status, and resolving order-related issues.

* Scope: My job is to help customers track their orders, provide updates on delivery status, and assist in resolving any issues related to their orders. I should not handle inquiries unrelated to orders, such as account management or product recommendations.

Instructions

The following instructions are used to run this subagent.

* Instruction: If a customer provides an order ID, use it to locate the order and provide the latest status.

Back ✓ ● ● Next

Step 3 — Add Topic 2 — Warranty Claims

1. Click **Add Topic** | Label: **Warranty Claims**

2. Paste the instructions below → **Save**

When a customer asks about a warranty claim or eligibility:

1. Ask for the product name, model, and original purchase date
2. Check if the product is within its warranty period (standard: 1 year from purchase)
3. If eligible, guide the customer to provide issue description and initiate a Warranty Claim
4. If a claim already exists, retrieve its current status and share it clearly
5. For any claim amount above 500, flag for manager approval before proceeding

Step 4 — Add Topic 3 — Product Support

1. Click **Add Topic** | Label: **Product Support**
2. Paste the instructions below → Save

When a customer reports a product issue or asks for troubleshooting:

1. Ask for the product name, model number, and description of the problem
2. Provide step-by-step troubleshooting guidance for common issues
3. If the issue cannot be resolved, create a Case with Case Type = Product Issue
4. Set priority: billing/data loss = High, performance = Medium, cosmetic = Low
5. Send the customer their case number and expected response time

Step 5 — Add Topic 4 — Case Status

1. Click **Add Topic** | Label: **Case Status**
2. Paste the instructions below → Save
3. Once all 4 topics are added, click **Activate** on the agent

When a customer asks about an existing support case:

1. Ask for their case number (format: 00001234)
2. Retrieve the case: status, priority, owner, last activity date
3. Share the current status in plain language (avoid internal jargon)
4. If the case has been open for more than 48 hours without update, escalate and notify the agent
5. Offer to add any new information the customer provides to the case notes

Developer Edition

Agentforce Builder

Customer Support Agent

Version 1

Settings

Help

Batch Test

Activate

Subagents

Manage the subagents assigned to your agent. To make changes, your agent must be deactivated.

New

Search subagents...

12 items • Sorted by Subagent Label(asc)

Subagent Label

Account Management

Case Management

Case Status

Delivery Issues

Escalation

General FAQ

Order Inquiries

Order Tracking

Product Support

Product Warranties and Claims

Reservation Management

Service Customer Verification

Session Started: Jun 27, 2026, 01:39 PM GMT

Session ID: 01...

"Can you help me..." Plan ID: f62417e9bbd8

User Prompt

Can you help me track my order?

Reasoning: Subagent Selecti

Prompt Used

Subagent Selected: Ser...

Verifies the customer's identity before granting access to sensitive data. Actions like managing cases, retrieving/managing orders, or making a reservation require identity verification. Sensitive data includes confidential, private, or security-protected information, such as business-critical data or personally identifiable information (PII). After verification is successful, don't repeat the process within the same

Conversation Preview

Hi, I'm an AI service assistant. How can I help you?

Can you help me track my order?

Sure! Before I can assist you with tracking your order, I need to verify your identity. Could you please provide your email address or username?

Describe your task or ask a question...

Prompt Builder — Case Resolution Summary Template

Step 6 — Create the Prompt Builder Template

- Setup → Quick Find → **Prompt Builder** → click **New Prompt Template**
- Template Type: **Flex Template** | Name: **Case Resolution Summary**

New Prompt Template

About

Select a prompt template type. The template type determines which data sources you can use in your prompts. [Learn more in Help.](#)

* Prompt Template Type

Flex

* Prompt Template Name

Case Resolution Summary

* API Name

Case_Resolution_Summary

Template Description

Enter a template description...

Inputs (Optional)

Add up to 5 data sources to include with your prompt instructions.

Cancel

Next

3. Associated Object: **Case**

4. Paste the prompt text below into the template editor

5. Use the merge field picker to insert the Salesforce fields (or paste directly)

6. Click **Save** → click **Activate**

You are a customer support AI for an Electronics & Gadgets company.

Review the case and provide a 2-sentence summary and 3 resolution steps for the agent.

Customer: `{!$Input:Case.Contact.Name}`

Case Type: `{!$Input:Case.Case_Type__c}`

Subject: `{!$Input:Case.Subject}`

Description: `{!$Input:Case.Description}`

Priority: `{!$Input:Case.Priority}`

Order Number: `{!$Input:Case.Order_Number__c}`

Format your response as:

SUMMARY: (2 sentences)

SUGGESTED STEPS: 1. 2. 3.

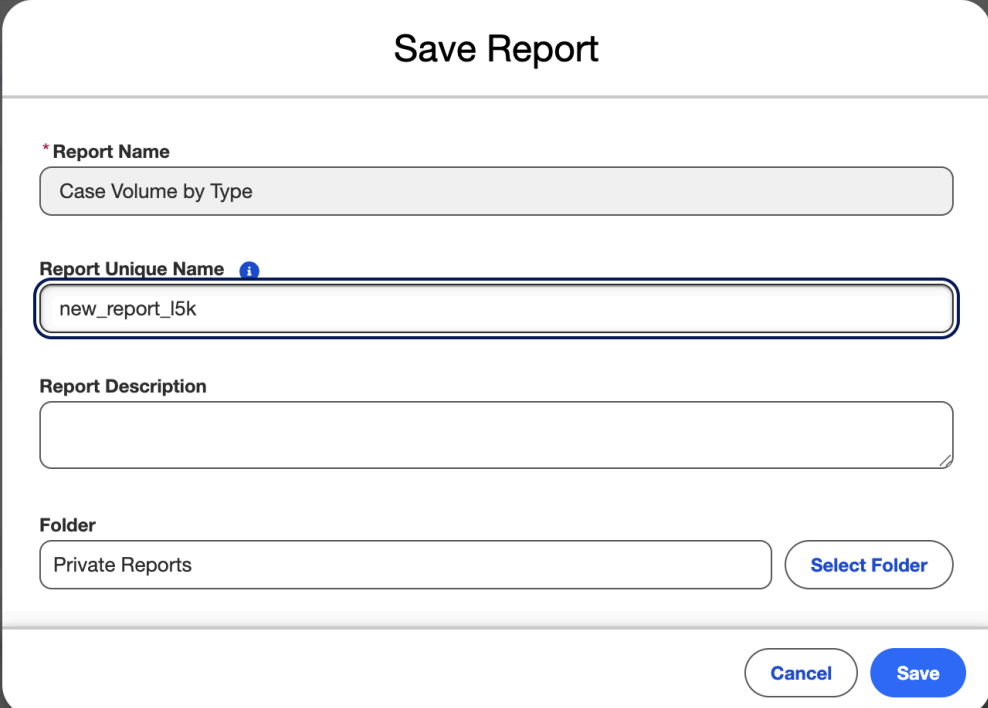
The screenshot shows the Salesforce Prompt Builder interface. At the top, there's a navigation bar with 'Setup', 'Home', and 'Object Manager'. Below that, a search bar and a 'Developer Edition' badge are visible. The main header area includes a 'Prompt Builder' tab and a 'Case Resolution Summary' dropdown. The interface is divided into two main sections: 'Template Details' on the left and 'Prompt' on the right. The 'Template Details' section has a 'Properties' tab and a 'Resources' tab. Under 'Resources', there's a list of fields: 'Case ID' and 'Case Number'. The 'Prompt' section shows the prompt text: 'You are a customer support AI for an Electronics company. Review the case and suggest 3 resolution steps.' Below this, there are merge fields for 'Customer', 'Type', 'Subject', and 'Description'. The 'Format' is set to 'SUMMARY: (2 sentences)' and the 'STEPS' are '1. 2. 3.'.

Phase 7 Reports & Dashboard

Create 5 reports + 1 operations dashboard

■ ~20 min

All reports use Report Type: **Cases**. Path: **App Launcher** → **Reports** → **New Report** → **Cases**



A screenshot of a 'Save Report' dialog box. The dialog has a title bar with a close button (X) in the top right corner. The main content area contains four input fields: 'Report Name' (with a red asterisk), 'Report Unique Name' (with a blue information icon), 'Report Description', and 'Folder'. The 'Report Name' field contains 'Case Volume by Type'. The 'Report Unique Name' field contains 'new_report_l5k'. The 'Report Description' field is empty. The 'Folder' field contains 'Private Reports' and has a 'Select Folder' button to its right. At the bottom right of the dialog are 'Cancel' and 'Save' buttons.

Save Report

* Report Name
Case Volume by Type

Report Unique Name ⓘ
new_report_l5k

Report Description

Folder
Private Reports [Select Folder](#)

[Cancel](#) [Save](#)

Step 2 — Report 2 — Open Cases by Queue

1. New Report → Cases → Start Report
2. Add Filter: **Status** → **Not Equal To** → **Closed**
3. Group Rows by **Queue** | Chart: **Donut**
4. Save & Run → Name: **Open Cases by Queue**

Step 3 — Report 3 — Average Resolution Time by Queue

1. New Report → Cases → Start Report
2. Add Filter: **Status Equals Closed**
3. Group Rows by **Queue**
4. Add Summary: click **Age** column → select **Average**
5. Save & Run → Name: **Avg Resolution Time by Queue**

Step 4 — Report 4 — Escalated Cases This Week

1. New Report → Cases → Start Report

2. Add Filter: **Is Escalated Equals True**
3. Add Filter: **Created Date = Last 7 Days**
4. Leave as tabular (no grouping needed — it's a list)
5. Save & Run → Name: **Escalated Cases This Week**

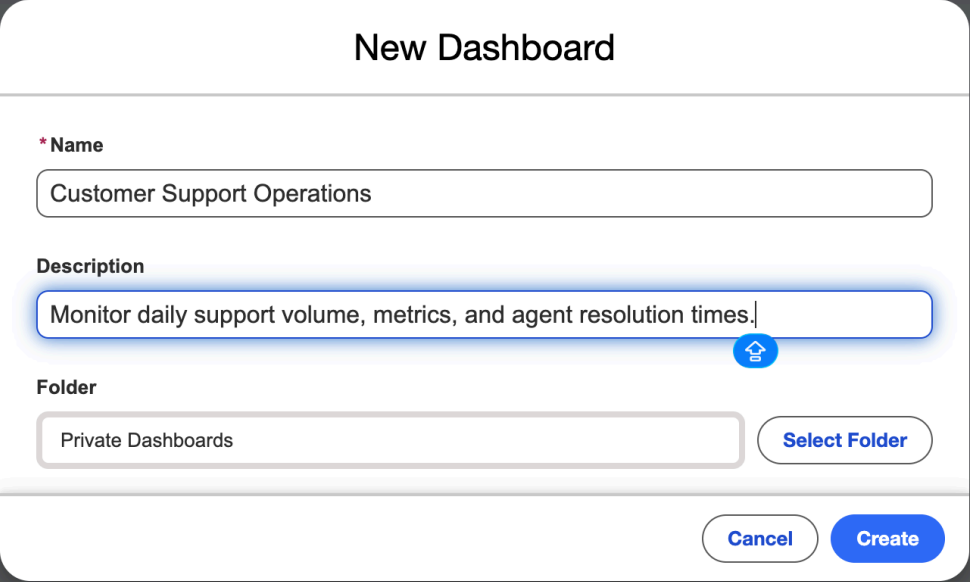
Step 5 — Report 5 — Cases Resolved Per Agent

1. New Report → Cases → Start Report
2. Add Filter: **Status Equals Closed** | Date: **This Month**
3. Group Rows by **Case Owner**
4. Add Summary: **Count** of cases
5. Save & Run → Name: **Cases Resolved Per Agent This Month**

SmartBridge SkillWallet | Salesforce Certified Administrator Program Page 17AI-Powered Customer Support & Service Automation Salesforce Project — Step-by-Step Guide

Step 6 — Build the Dashboard

1. App Launcher → **Dashboards** → click **New Dashboard**
2. Name: **Customer Support Operations** → Create



New Dashboard

*** Name**

Customer Support Operations

Description

Monitor daily support volume, metrics, and agent resolution times.

Folder

Private Dashboards [Select Folder](#)

[Cancel](#) [Create](#)

3. Click **+ Component** → select **Case Volume by Type** → choose **Bar Chart** → Add

Add Widget

☐ Use chart settings from report

☐ Use table settings from report

Display As















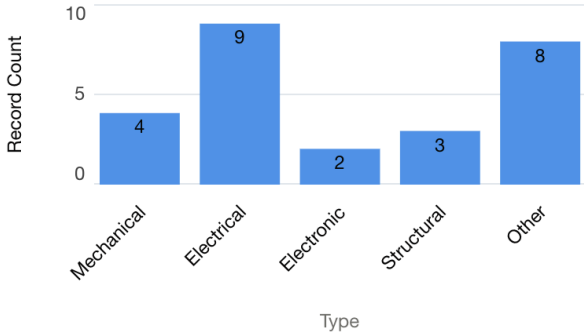
X-Axis

Y-Axis

☐ Show Reference Line

Preview

Case Volume by Type



View Report (Case Volume by Type)

Cancel
Add

4. Click **+ Component** → select **Open Cases by Queue** → choose **Donut** → Add
5. Click **+ Component** → select **Escalated Cases This Week** → choose **Table** → Add
6. Click **+ Component** → select **Avg Resolution Time by Queue** → Add
7. Click **+ Component** → select **Cases Resolved Per Agent** → Add
8. Arrange the 5 components in a 2x2 grid with the escalated cases table spanning the bottom
9. Click **Save** → click **Subscribe** (optional) → share with all internal users



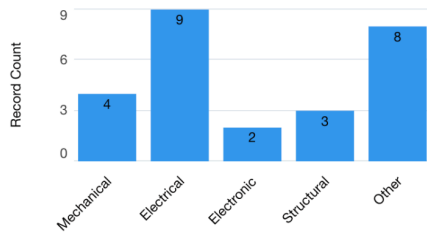
Q Search...



Developer Edition

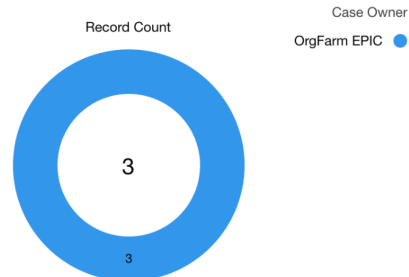
Welcome

Case Volume by Type



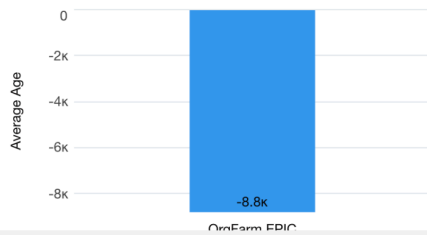
[View Report \(Case Volume by Ty...](#) As of Jun 27, 2026, 3:11 AM

Open Cases by Queue

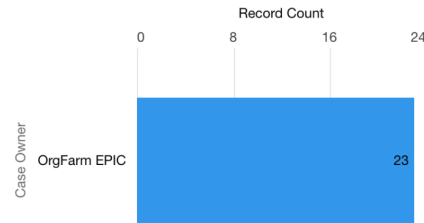


[View Report \(Open Cases by Qu...](#) As of Jun 27, 2026, 3:11 AM

Avg Resolution Time by Queue



Cases Resolved Per Agent



7.TESTING AND VALIDATION

7.1 Introduction

Testing is a critical phase in the Software Development Life Cycle (SDLC) that ensures the developed application functions correctly and meets all specified requirements. It helps identify defects, validate business logic, verify automation, and ensure the reliability of the system before deployment.

For the AI-Powered Customer Support & Service Automation system, testing was performed on all major modules, including customer management, case management, warranty claims, Salesforce Flows, queues, Agentforce, Prompt Builder, reports, and dashboards. Both functional and integration testing were carried out to verify that each component worked as expected.

7.2 Testing Objectives

The primary objectives of testing were:

- Verify that all functional requirements are implemented correctly.
- Ensure Salesforce Flows execute without errors.
- Validate data entered into the system.
- Verify queue-based case assignment.
- Test Agentforce AI responses.
- Validate Prompt Builder templates.
- Ensure reports and dashboards display accurate information.
- Verify overall system reliability and performance.



New Case

* = Required Information

Case Information

Case Owner

 Bhanu sri Inturi

*** Status**

New ▼

Case Number

Priority

Medium ▼

Contact Name

Search Contacts... 

*** Case Origin**

--None-- ▼

Account Name

Search Accounts... 

Type



Other ▼

☐ Send notification email
to contact...

--None-- ▼

Cancel

Save & New

Save

Developer Edition

Welcome

Case

+ Follow

Edit

Delete

Change Owner

Priority

Medium

Status

New

Case Number

00001026

Case Owner

Product Issues Queue

Case Number

00001026

Contact Name

Account Name

Type

Other

Case Reason

Status

New

Priority

Medium

Contact Phone

Contact Email

Case Origin

Web

Warranty Claims (0)

Filters: All time • All activities • All

Refresh • Expand All

Upcoming & Overdue

No activities to show.

Get started by sending an email, scheduling a task, ar

7.3 Types of Testing Performed

Testing Type	Purpose
Unit Testing	Test individual modules independently.
Functional Testing	Verify business functionality.
Integration Testing	Ensure different modules work together.
Validation Testing	Validate user inputs and business rules.
User Acceptance Testing	Confirm the application meets user expectations.

7.4 Test Cases

Test Case ID	Test Scenario	Expected Result	Status
TC-01	Create Customer Case	Case created successfully	Passed

TC-02	Queue Assignment	Case assigned to correct queue	Passed
TC-03	Warranty Claim Submission	Warranty record created	Passed
TC-04	Approval Process	Manager receives approval request	Passed
TC-05	Flow Automation	Email notification sent automatically	Passed
TC-06	Agentforce Response	AI provides relevant suggestions	Passed
TC-07	Report Generation	Report displays accurate records	Passed
TC-08	Dashboard	Dashboard updates automatically	Passed

7.5 Validation Results

The application was tested under various scenarios, and all major modules performed successfully. Validation rules prevented incorrect data entry, flows executed without failures, AI responses were generated correctly, and reports reflected real-time information.

8 RESULTS AND DISCUSSION

8.1 Introduction

This chapter presents the outcomes of implementing the AI-Powered Customer Support & Service Automation system. The developed application successfully automates customer support operations, improves service efficiency, and provides AI-assisted customer interactions using Salesforce Service Cloud and Agentforce.

8.2 Project Outcomes

The developed system successfully achieved the following:

- Automated customer case creation.
 - Intelligent case assignment using queues.
 - Warranty claim management.
 - Automated email notifications.
 - AI-powered customer assistance.
 - Real-time reports and dashboards.
 - Reduced manual intervention.
 - Improved service quality.
-

8.3 Performance Improvements

Existing System	Proposed System
Manual case assignment	Automatic queue assignment
Delayed responses	Instant acknowledgements
Manual reporting	Real-time dashboards
Repetitive support tasks	Automated Salesforce Flows
No AI assistance	Agentforce AI recommendations

8.4 Benefits Achieved

Customer Benefits

- Faster response times
- Improved service quality
- Easy order tracking
- Transparent warranty process

Business Benefits

- Reduced operational costs
- Higher employee productivity
- Better SLA compliance
- Improved customer satisfaction

Technical Benefits

- Cloud-based architecture
 - Scalable platform
 - Secure data management
 - Low-code customization
-

CHAPTER 9

ADVANTAGES, LIMITATIONS AND FUTURE ENHANCEMENTS

9.1 Advantages

The AI-Powered Customer Support & Service Automation system offers numerous advantages:

- Centralized customer support platform.
 - Automated case management.
 - AI-assisted customer interactions.
 - Reduced manual workload.
 - Faster response and resolution time.
 - Automated notifications.
 - Efficient warranty claim management.
 - Real-time reports and dashboards.
 - Improved customer satisfaction.
 - Highly scalable cloud infrastructure.
-

9.2 Limitations

Although the application meets the project objectives, certain limitations exist:

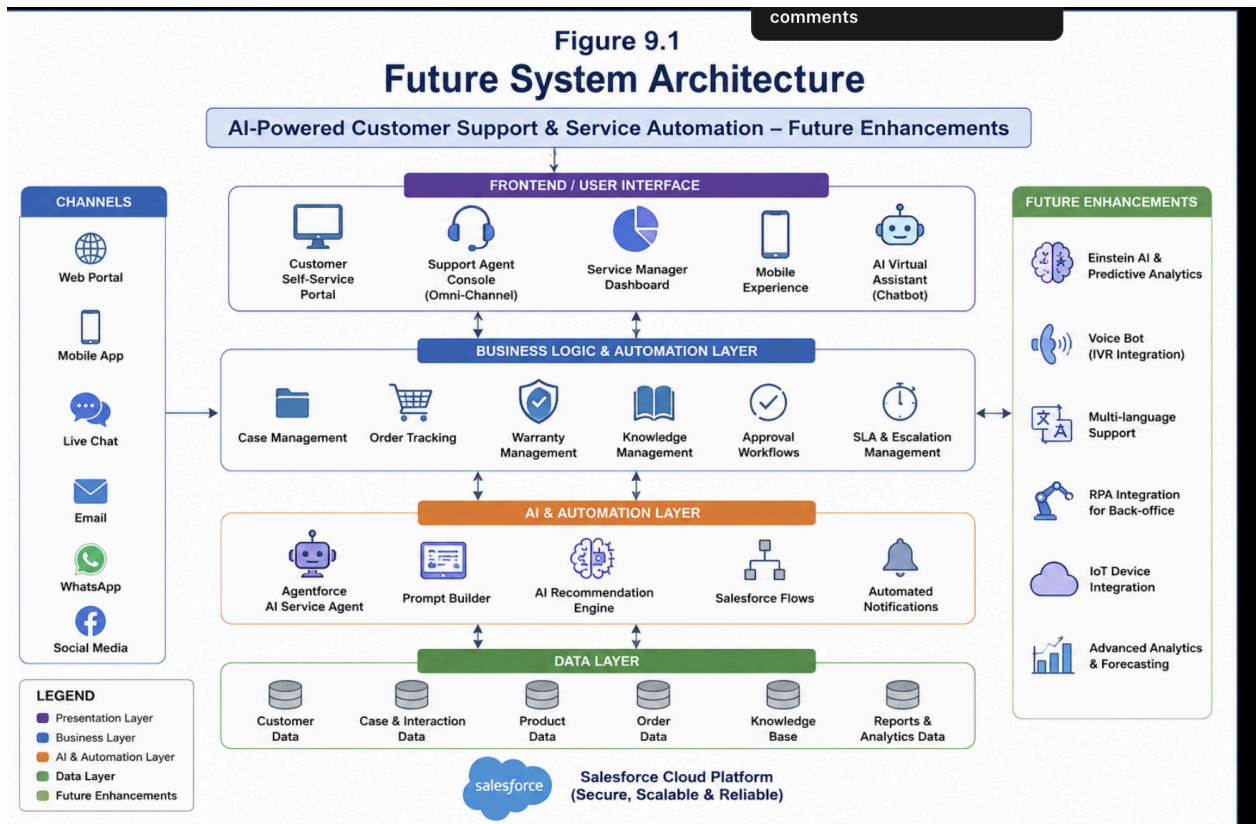
- Internet connectivity is required.
 - Some advanced Salesforce features require premium licensing.
 - AI responses depend on configured prompts and knowledge sources.
 - Integration with external systems was not implemented in this project.
 - Live chat and voice support are outside the current project scope.
-

9.3 Future Enhancements

The system can be enhanced by integrating additional technologies and features such as:

- WhatsApp Business integration.

- Live Chat support.
- Voice-enabled AI assistant.
- Multi-language customer support.
- Predictive case routing using AI.
- Customer satisfaction surveys.
- IoT device monitoring for smart electronics.
- Mobile application integration.
- AI-based sentiment analysis.
- Integration with ERP systems.



9.4 Chapter Summary

The developed application provides a strong foundation for AI-powered customer service automation. Future enhancements can further improve customer engagement, intelligent automation, and business scalability.

CHAPTER 10

CONCLUSION

10.1 Conclusion

The **AI-Powered Customer Support & Service Automation** system was successfully designed and implemented using **Salesforce Service Cloud, Salesforce Flow, Agentforce, and Prompt Builder**. The application addresses the challenges of traditional manual customer support by automating repetitive business processes, improving response times, and providing intelligent assistance through AI.

The developed solution effectively manages customers, products, orders, support cases, warranty claims, queues, approval processes, and AI-generated responses within a centralized Salesforce environment. Automated workflows significantly reduce manual effort while ensuring consistency, accuracy, and timely communication with customers.

Agentforce enhances the support process by providing contextual recommendations and intelligent responses, enabling support agents to resolve customer issues more efficiently. Real-time reports and dashboards provide managers with valuable insights into service performance, case resolution, and workload distribution.

Overall, the project demonstrates how Salesforce's low-code platform can be leveraged to build scalable, secure, and intelligent customer support solutions. The implementation not only improves operational efficiency but also enhances customer satisfaction and prepares the organization for future business growth.

10.2 Project Achievements

The project successfully achieved all major objectives:

- Automated customer support workflows.
- Intelligent queue-based case assignment.
- Warranty claim automation.
- AI-powered customer assistance.
- Salesforce Flow automation.
- Prompt Builder integration.
- Interactive reports and dashboards.
- Secure and scalable Salesforce implementation

11.REFERENCES

The following references were used during the development of the AI-Powered Customer Support & Service Automation project:

1. Salesforce. *Trailhead Learning Platform*.
2. SmartBridge. *Salesforce SkillWallet Learning Materials*.
3. Roger S. Pressman. *Software Engineering: A Practitioner's Approach*. McGraw-Hill.
4. Ian Sommerville. *Software Engineering*. Pearson Education.
5. Agile Alliance. *Agile Software Development Guide*.